MODULE 1

Introduction to Renewable Energy Sources

Course Objectives

To understand energy scenario, energy sources and their utilisation, renewable energy tecchnologies and conversion, presend needs and future demands of society, energy conservation.

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the environmental aspects of renewable energy resources in comparison with various conventional energy systems, their prospects and limitations.	**K1**
CO2	Understand the concepts of wind energy.	**K1**
CO3	Describe the use of solar energy and the various components used in the energy production with respect to applications like-heating, cooling, desalination, power generation.	**K2**
CO4	Understand the concept of biomass energy resources and conversion principles of tidal energy.	**K2**
CO5	Acquire the basic knowledge of ocean thermal energy conversion. Understand the principle of green energy and hydrogen energy.	**K1**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	3										2
CO3	3	3										2
CO4	3	3										2
CO5	3	3										2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Non-conventional energy sources	G. D. Rai	Khanna	4th edition 2023
2	Renewable energy systems	Thomas E. Kissell, David M. Buchla, Thomas L. Floyd,	Pearson	2017
3	Non-Conventional Energy Resources	Sawhney G. S.	PHI Learning	2012
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End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction:** Principles of renewable energy; energy and sustainable development, fundamentals and social implications. Worldwide renewable energy availability, renewable energy availability in India, types of renewable energy. **Wind Energy:** Properties of wind, availability of wind energy in India, wind velocity and power from wind (numerical problems); major problems associated with wind power, Basic components of wind energy conversion system (WECS); Classification of WECS- Horizontal axis- single, double and multi-blade system. Vertical axis - Savonius and Darrieus types.	9

ENERGY RESOURCE

- An energy resource is something that can produce heat, power life, move objects, or produce electricity.
- Matter that stores energy is called a fuel.
- Energy resources are all forms of fuels used in the modern world, either for heating, generation of electrical energy, or for other forms of energy conversion processes.
- Energy resources can be roughly classified in three categories: fossil, nuclear and renewable.

Classification of Energy Resources

1. Based on Usability of Energy
   a) Primary Resources
      - Energy extracted or captured directly from natural resources
      - Example: Coal, Oil, Gas, etc.



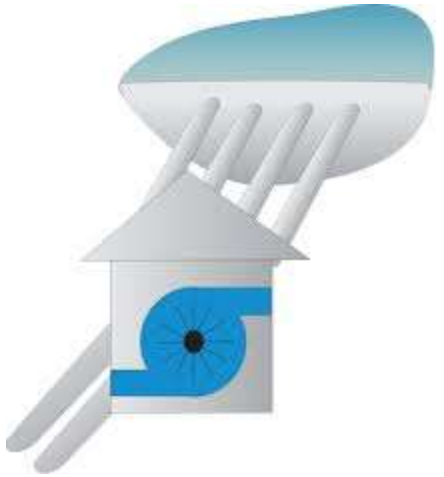


   b) Secondary Resources
      - Transformed from primary resources
      - Example: Steam, Electricity, etc.

Classification of Energy Resources

2. Based on Traditional Use
   a) Conventional
      - Being traditionally used since decades
      - Example: Fossil fuels, nuclear, hydro etc.







   b) Non-conventional
      - Large-scale use after oil crisis (1973)
      - Example: Solar, Wind, etc.





Classification of Energy Resources

3. Based on Long-Term Availability
   a) Non-renewable
      - Finite natural resources
      - Does not get replenished at the rate of consumption
      - Example: Fossil fuels, Uranium, etc.
   b) Renewable
      - Obtained from natural resources
      - Essentially inexhaustible
      - Example: Wind power, Solar power, etc.

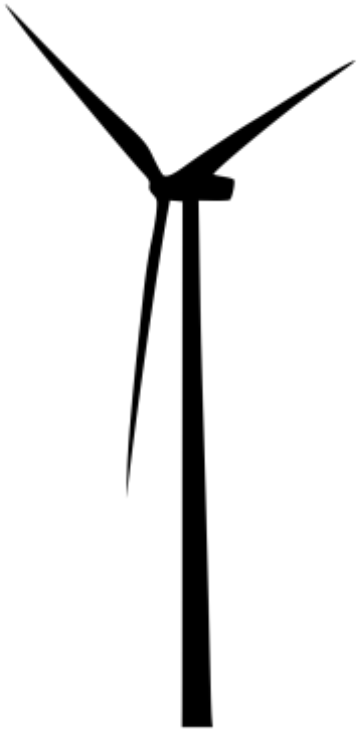
Classification of Energy Resources

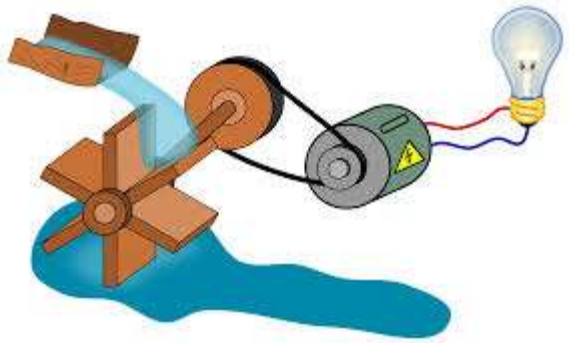
4. Based on Commercial Application
   a) Commercial Energy
      - Available in the market for a definite price
      - Example: Electricity, Diesel, etc.
   b) Non-commercial Energy
      - Not available in the commercial market
      - Typically used in rural households
      - Example: Firewood, Solar energy for water heating etc.

Classification of Energy Resources

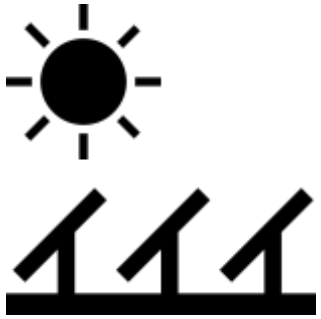
5. Based on Origin
   - Fossil fuels energy
   - Nuclear energy
   - Hydro energy
   - Solar energy
   - Wind energy
   - Biomass energy
   - Geothermal energy
   - Tidal energy
   - Ocean Thermal energy
   - Ocean Wave energy













Conventional Energy Sources
a)Fossil Fuel

Fossil Fuels are non-renewable sources of energy, which, if exhausted, can not be replenished in a short time.

- Their reserves are limited and are considered very precious.
- These are also contributing to the global environmental pollution.

Fossil Fuels ???

Fossil fuels are hydrocarbons, primarily coal, fuel oil or natural gas, formed from the remains of dead plants and animals.

Example ???



Coal **Oil** **Gas**

b)Coal

- Since the advent of industrialization coal has been most common source of energy.
- In the last three decades, the world switched over from coal to oil as a Major source of energy because it is simpler and cleaner to obtain useful energy from oil.
- Coal is a complex mixture of compounds of carbon, hydrogen and oxygen.
- Small amounts of nitrogen and Sulphur compounds are also present in coal.
- It is mainly available in Bihar, West Bengal, Orissa and Madhya Pradesh.
- The big coal mines in our country are at Jharia and Bokaro in Bihar and at Raniganj in West Bengal.
- It is considered as the backbone of the energy sector for its use in industry, transportation and electric power generation.

Petroleum or Oil

- It is a dark coloured, viscous and foul smelling crude oil.
- The petroleum means rock oil.
- It is normally found under the crust of earth trapped in rocks.
- The crude oil is a complex mixture of several solid liquid gaseous hydrocarbons mixed with water, salt and earth particles.
- It is a natural product obtained from oil wells.

Some of the crude oil producing locations in our country are:

- Ankleshwar and Kalol in Gujarat
- Rudrasagar and Lakwa in Assam; and
- Bombay high (off-shore area)

Natural Gas

- It consists about 95% Methane and rest ethane and propane.
- It occurs deep under the crust of the earth either alone or a long with oil above the petroleum deposits.
- It is a product of petroleum mining.
- The gas is available in Tripura, Jaisalmer, off-shore areas of Bombay High and in the Krishna – Godavari delta.
- It is used as a domestic and industrial fuel.
- The natural gas is now also available as CNG (Compressed Natural Gas) a substitution of petrol in automobiles.

Conventional Power Generation
Thermal Power

- **Thermal power** is the "largest" source of power in **India**.
- There are three types of power plants-

Steam Turbine Power Plants-use coal ,oil ,gas as fuel

Gas Turbine Power Plants-oil or gas as fuel

Diesel power Plants-Prime Mover is an IC Engine and diesel oil used as fuel



The **Vindhyachal Thermal Power Station** in the Singrauli district of Madhya Pradesh, with an installed capacity of 4,760MW, is currently the biggest thermal power plant in India. It is a coal-based power plant owned and operated by NTPC

Thermal Power Plants in Kerala

Name	Location	Capacity in MW	Year Opened
Brahmapuram Diesel Power Plant	Kochi	106.6	1997
Kozhikode Diesel Power Project	Nallalam	128	2000
Rajiv Gandhi Combined Cycle Power Plant(NTPC)	Kayamkulam	359.58	1998

Advantages

- Availability of Power is independent of weather conditions
- Can be installed at any place where

Disadvantages

- High Maintanence and operating cost
- Polluting Source
- Ash disposal difficulty
- Long start up time

Hydro Electric Power

- Potential energy of water is converted into Mechanical energy by using prime movers known as hydraulic turbines.
- Water power is quite cheap where water is available in abundance.
- Although capital cost of hydro electric power plants is higher as compared to other types of power plants but their operating Costs are quite low, as no fuel is required in this case

Hydo Electric Power Generation



The Three Gorges Dam in Hubei, China, has the world's largest instantaneous generating capacity (22,500 MW).



Itaipu Dam in Brazil/Paraguay in second place (14,000 MW).

Despite the large difference in installed capacity, these two power stations generate nearly equal amounts of electrical energy during the course of an entire year due to much variance in seasonal flow of Three Gorges dam

- India is the seventh largest producer of Hydroelectric energy in the world.

Name	River	Location	Capacity (MW)
Tehri Dam	Bhagirathi	Uttarakhand	2400 MW
Srisailam Dam	Krishna	Andhra Pradesh	1670 MW
Nagarjunasagar	Krishna	Andhra Pradesh	965 MW
Sardar Sarovar	Narmada	Gujarat	1450 MW

Tehri dam is the highest Dam in India with a height of 260.5 metres. It is also listed in the top ten highest dams in the world.

In Kerala

- **Idukki Hydro Electric Project** (IHEP) is the largest hydro **electric project** in Kerala. Installed capacity of this project is 780 MW. This is the first arch dam constructed in India. This is a double curvature parabolic arch dam, constructed across the river Periyar. With a height of 168.91 meters, this is a second highest dam in India.

Advantages

- Low Start up time
- No Pollutants
- Longer Life

Disadvantages

- Large area required
- May affect ecological balance of the area
- High initial cost
- Long gestation period
- Weather dependant

Nuclear Power

- The heat produced by nuclear fission of the atoms of fissionable material is utilized in special heat exchangers for the production of steam which is then used to drive turbo generators as in the conventional power plants.

Largest nuclear power station of world, the Kashiwazaki-Kariwa (Japan) at 7,965 MW



Nuclear Power In India

Power station	Operator	State	Type	Units	Total capacity (MW)
Kudankulam[116]	NPCIL	Tamil Nadu	VVER-1000	1000 × 2	2,000
Madras (Kalpakkam)	NPCIL	Tamil Nadu	PHWR	220 × 2	440
Narora	NPCIL	Uttar Pradesh	PHWR	220 × 2	440
Rajasthan	NPCIL	Rajasthan	PHWR	100 × 1 200 x 1 220 × 4	1,080
Tarapur	NPCIL	Maharashtra	BWR PHWR	160 x 2 540 × 2	1,400
Total					7,380



Kudankulam Nuclear Power Plant is located in the Tamil Nadu, Southern India. It is the highest-capacity nuclear plant in India, with a total of 2,000MW currently installed with a further 2,000MW under construction.

Advantages

- Less area required
- Less fuel required
- Independent of weather conditions

Disadvantages

- High Initial Cost
- Radiation hazard
- Expensive Fuel
- Waste Disposal

Significance of Non Conventional Resources

- Depletion of fossil fuels
- Increase in energy demand
- Pollution by fossil fuels-due to burning of coal ,major pollutants are emitted.

  CO,CO2-Causes global warming

  Hydrocarbons-damages ozone layer

  Sulphur dioxide,Nitrogen oxides –respiratory problems
- Projects like hydro power affect wildlife , cause deforestation
- Radiation hazards and disposal of nuclear waste in the case of nuclear power plants.

The sources of energy which are being produced continuously in nature and are in exhaustible are called renewable sources of energy (or) non-conventional energy.

Some of these sources are:

1. Solar energy
2. Wind energy
3. Bio mass
4. Tidal energy
5. Wave energy
6. Geo thermal

COMPARISON OF CONVENTIONAL AND NON CONVENTIONAL SOURCES

Conventional sources of energy	Non-conventional sources of energy
These sources of energy are not abundant, present in limited quantity, e.g. coal, petroleum, natural gas.	These sources of energy are abundant in nature, e.g. solar energy, wind energy, tidal energy, biogas from biomass etc.
They have been in use for a long time.	They are yet in development phase over the past few years.
They are not replenished continuously. They are formed over a million years.	They are replenished continuously by natural processes.
They are called non-renewable sources of energy.	They are called renewable sources of energy.
They can be exhausted completely due to over-consumption except for hydel power.	They cannot be exhausted completely.
They pollute the environment by emitting harmful gases and also contribute to global warming.	They are environment-friendly, do not pollute the environment.
They are commonly used for industrial and commercial purposes.	They are used commonly used for household purposes.
Heavy expenditure is involved in using and maintaining these sources of energy.	Using these sources is less expensive.
They are used extensively, at a higher rate than the non-conventional sources.	They are not used as extensively as conventional sources.

Principles of Renewable Energy

- The principles of renewable energy are based on utilizing natural, self-replenishing sources like the sun, wind, and water, which are not depleted with use.
- Unlike fossil fuels, these sources are constantly renewed on a human timescale and have a lower environmental impact, often with zero or low carbon emissions.
- Key principles include harnessing naturally occurring processes (like solar or wind) in the local environment, converting energy through different processes (mechanical, heat, or photon), and implementing systems that are often dispersed rather than centralized.

Sustainable energy

- Affordable
- Reliable
- Clean & low carbon
- Efficiently used
- Accessible to all

Sustainable Energy Development

- Renewable Energy Sources- Solar, Wind, Biomass, Ocean Energy etc
- Energy Efficiency-Use of LED lighting, efficient motors, insulated buildings, smart grids, efficient appliances
- Energy Access for All
- Environmental Protection
- Policy & Governance- Policies supporting: Renewable energy investments,Carbon pricing,Green energy subsidies,Research and innovation
- Challenges-High initial cost of renewable systems,Energy storage limitations,Grid integration issues

1	No poverty
2	Zero hunger
3	Good health and well-being
4	Quality Education
5	Gender equality
6	Clean water and sanitation
7	Affordable and clean energy
8	Decent work and economic growth
9	Industry, innovation and infrastructure
10	Reduced inequalities
11	Sustainable cities and economies
12	Responsible consumption and production
13	Climate action
14	Life below water
15	Life on land
16	Peace, justice and strong institutions
17	Partnership for the goals

Sustainable Development Goals (SDGs)

1	No poverty
2	Zero hunger
3	Good health and well-being
4	Quality Education
5	Gender equality
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- 10/12/25 4th
- 1,12,13,14,25,27,40,56,61,65,67,68,71

World wide renewable energy

- Renewable energy is globally abundant, widely available in every country, and rapidly expanding, already supplying over 30% of global electricity (mostly hydro, wind, solar) and set to become the largest power source by 2025-2026 driven by falling costs and huge capacity additions, especially in China, the U.S., and India, though faster deployment is needed to meet climate goals.

- Solar, wind, hydro, geothermal, and biomass resources exist in all nations, reducing reliance on imported fossil fuels.

- Solar and wind are now the most affordable new electricity sources in most regions, with falling prices making them highly competitive.
- Renewables are expected to supply nearly 45% of global electricity by 2030
- Worldwide solar installed capacity reached over 2.25 Terawatts (TW) by late 2024.

Solar Energy –Impact on environment

- Solar energy is a very large, inexhaustible source of energy.
- Solar energy is received in the form of radiation,can be converted directly or indirectly into other forms of energy, such as heat & electricity.
- High Production Cost
- Large Scale electricity generation requires large amount of land
- Solar PV modules pose disposal problem due to presence of arsenic and cadmium
- Hazards to eyesight from solar reflectors

- 12/12/25 1st hr
- 1,5,25,33,51,58,65,68

Major Solar Power Producers

- China: 890 GW (dominating global capacity).
- USA: 236 GW (driven by policies like the IRA).
- India: 119 GW (rapid growth from rooftop and utility-scale projects).
- Germany: 109 GW (strong rooftop solar adoption).
- Japan: 100 GW (focus on floating solar).
- The largest solar power plant in the world is the Bhadla Solar Park in Rajasthan, India, with an installed capacity of 2,245 MW. It covers approximately 14,000 acres.

Major Solar Power Plants In India

1. Bhadla Solar Park(Rajasthan) – 2,250MW(Largest solar power plant in the world).
2. Shakti Sthala solar power project(Karnataka) – 2,050MW
3. Ultra Mega Solar Park(Andhra Pradesh) – 1,000MW
4. Rewa Solar Power Project (Madhya Pradesh)– 750MW
5. Kamuthi solar power plant (Tamil Nadu)– 648MW

Wind Energy

- Energy of wind can be economically used for the generation of electricity.
- Winds are caused from 2 main factors:

1.Heating & cooling of the atmosphere which generates convection currents. Heating is caused by the absorption of solar energy on the Earth‘s surface & in the atmosphere.

2. The rotation of the Earth with respect to atmosphere & its motion around the sun

Impact on Environment

- Large area required
- Noise Pollution
- Hazards for birds
- Interference with television siganls

Top 5 Countries in wind power

1.China – installed capacity 560 GW

2. US – installed capacity 154GW

3. Germany – installed capacity 70W

4. India – installed capacit 50GW

5. Brazil,Spain – installed capacity 32GW

Worldwide wind installed capacity surpassed 1.1 Terawatts (TW) by the end of 2024

Wind Power in India

- Muppandal, Tamil Nadu: Largest onshore farm (1500 MW).
- Jaisalmer, Rajasthan: Second largest (1064 MW).
- Brahmanvel, Maharashtra: 528 MW.
- Dhalgaon, Maharashtra: 278 MW.
- Vankusawade, Maharashtra: 259 MW.
- Poolavadi, Tamil Nadu: 99 MW.

Major States for Wind Energy

- Leading in Capacity: Gujarat and Tamil Nadu.
- High Potential (120m height): Rajasthan, Gujarat, Maharashtra, Andhra Pradesh, Karnataka.

Roll No	state	Wind Potential at 120 m (GW)	Wind Potential at 150 m (GW)
1	Andhra Pradesh	74.90	123.3
2	Gujarat	142.56	180.8
3	Karnataka	124.15	169.3
4	Madhya Pradesh	15.40	55.4
5	Maharashtra	98.21	173.9
6	Rajasthan	127.75	284.2
7	Tamil Nadu	68.75	95.1
8	Telangana	24.83	54.7

Power Plant	Location	State	Capacity(MW)	Producer
Kutch Wind Farm (Gujarat Hybrid Renewable Energy Park)	Kutch	Gujarat	11,500 (wind)+ 11,500 (solar + wind)	Adani Group[47] Suzlon[48]
Muppandal Wind Farm	Kanyakumari	Tamil Nadu	1500	Muppandal Wind
Jaisalmer Wind Park	Jaisalmer	Rajasthan	1064	Suzlon Energy
Brahmanvel windfarm	Dhule	Maharashtra	528	Parakh Agro Industries
Sidhpur-2	Dwarka	Gujarat	250.8	KP Energy
Kayathar	Tutcorin	Tamilnadu	300	Siemens Gamesa, ReNew Power
Dhalgaon windfarm	Sangli	Maharashtra	278	Gadre Marine Exports

Biomass

- Biomass is organic material from plants and animals used as a renewable energy source, including wood, crops, waste, and animal manure, that can be converted into heat, electricity, or biofuels, acting as a sustainable alternative to fossil fuels by recycling carbon in the atmosphere.
- Sources: Agricultural residues (straw, crop waste), forestry waste (wood), animal manure, municipal solid waste, energy crops, and sewage sludge.
- Biomass energy is considered carbon-neutral because plants absorb CO2 as they grow, releasing the same amount when used as fuel.

Conversion Methods:

- Direct Combustion: Burning biomass to heat water, create steam, and drive turbines for electricity.
- Gasification/Pyrolysis: Heating biomass to create syngas or bio-oil.
- Anaerobic Digestion: Microbes break down organic matter to produce biogas (methane).
- Fermentation: Converting plant sugars into ethanol.

Uses: Electricity generation, heating, and production of transportation fuels like biodiesel and bioethanol.

- China: World leader in total biopower capacity (approx. 31 GW in 2023) and production (25% of global biopower), using agricultural waste and waste-to-energy.
- United States (US): Major player, with ~11 GW capacity, and a top producer of ethanol.
- Brazil: Huge biopower presence (18 GW capacity), especially from sugarcane byproducts for biofuels (ethanol).
- India: Growing significantly, with ~11 GW capacity and major biofuel (ethanol/biodiesel) growth, becoming a top producer.
- Indonesia: Leading biodiesel producer (palm oil) and a significant contributor to biomass energy.

- Dominant Sources: China uses waste, Brazil uses sugarcane, US uses corn/soy, Indonesia uses palm oil
- Biopower vs. Biofuels: Some countries lead in electricity (China, US, Brazil), while others lead in liquid biofuels (US, Brazil, Indonesia).

Biomass power in India

- India has a significant biomass power sector with a cumulative installed capacity of over 11.5 GW as of March/October 2025, encompassing biomass power, bagasse/non-bagasse cogeneration, and waste-to-energy, with major contributions from states like Maharashtra, Karnataka, and Andhra Pradesh.

- Bagasse cogeneration is a process that uses the fibrous residue from sugarcane, called bagasse, to simultaneously generate electricity and steam.
- This is a highly efficient form of renewable energy, where the sugar mill uses the generated power and steam for its own needs and can export surplus electricity to the grid.
- It is a common practice in sugar-producing countries to make the mills self-sufficient and generate additional income from selling surplus power.
- Leading States: Maharashtra (approx. 2,907 MW), Uttar Pradesh (approx. 1,985 MW), Karnataka (approx. 1,869 MW).

Geothermal Energy

- Geothermal energy is the heat or steam that comes from the earth's subsurface. It is found in the rocks and fluids below the earth's crust and can be traced as far down to the core that contains hot molten rock, magma.
- In order to generate power from geothermal energy, a mile deep wells are dug into underground reservoirs to access the steam and hot water, which are used to drive turbines connected to electricity generators.

- **Huge Potential**
- **Geothermal Is a Reliable & Renewable Source of Energy**
- **Geothermal Systems are Highly Efficient**
- **Geothermal Systems Need Little or No Maintenance**
- **Greenhouse Emissions**
- **Geothermal System has High Investment Costs**
- **Geothermal Energy is Location-Specific**
- **Co2 emission less compared to thermal plants**

Impacts

- Gases escaping to atmosphere like H_2S are oxidized to SO_2 and H_2SO_4 and drop down as acid rain
- Noise pollution caused by exhausts,blow downs and centrifugal separation.Can be controlled by silencers
- Discharge of hot water to river affects drinking water , farming and fisheries
- Chemicals like sulphates, chlorides and carbonates of lead , boron and arsenic pollute soil and water
- Large scale withdrawal of underground fluids may trigger damage to surface structures.

Top Countries by Capacity (approx. 2024/2025)

- United States: Leading producer with the most installed capacity (around 3.9 GW).
- Indonesia: Second largest, with significant development.
- Philippines: A major player, ranking third globally.
- Turkey: Rapidly growing, placing fourth.
- New Zealand: Strong capacity, often above 1 GW.

- India's geothermal energy potential is substantial, estimated at over 10,000 MW, though the sector is still in its early stages.
- The country has identified around 340 hot springs, with the most promising sites for power generation including the Puga valley in Ladakh, which is set to host India's first geothermal power project.
- Other promising locations for geothermal power plants include Manikaran (Himachal Pradesh), Jalgaon (Maharashtra), Tapovan (Uttarakhand), Bakreshwar (West Bengal), and Tuwa (Gujarat).

Ocean Thermal Energy Conversion (OTEC)

- **Ocean Thermal Energy Conversion** (**OTEC**) uses the ocean thermal gradient between cooler deep and warmer shallow or surface seawaters to run a heat engine and produce useful work, usually in the form of electricity.

Impacts

- Leakage of ammonia used as working fluid may cause great damage to eco system
- Hazards to fishes
- An OTEC plant displaces nearly 4 cumec water per MW generation

This disturbs thermal balance ,salinity gradient and creaters adverse effect on marine environmnet

- United States (Hawaii) hosts the largest operational test/demonstration facilities (like Makai's 100kW plant at Natural Energy Laboratory of Hawaii Authority) and is a leader in OTEC development, with China, Japan, and India also heavily involved in research and pilot projects.

Current OTEC project in India

- Location: Kavaratti, Lakshadweep.
- Purpose: To power a desalination plant that will produce about 100,000 to 150,000 liters of potable water per day.
- Technology: The plant is based on the closed-cycle OTEC method, which uses a working fluid like ammonia to generate electricity by harnessing the temperature difference between surface and deep ocean waters.
- Power Generation: The OTEC plant is projected to generate between 60 and 65 kW of power.
- Significance: This is India's first OTEC plant and a pioneering effort to provide clean drinking water using a green energy source

- Sihwa Lake Tidal Power Station (South Korea): The world's largest, with 254 MW capacity, improving water quality in the lake.
- La Rance Tidal Power Station (France): The first and second-largest, operating since the 1960s with 240 MW.
- Annapolis Royal Generating Station (Canada): A significant plant in Nova Scotia, around 20 MW.
- Jiangxia Tidal Power Plant (China): Operational since the 1980s, with a smaller capacity (around 3.2 MW).

Tidal Power in India

- Gulf of Kutch, Gujarat: Considered the leading site, with plans for a 50 MW pilot project near Mandvi and a larger 200 MW project by Gujarat Power Corporation Ltd (GPCL) and Atlantis Resources.
- Durgaduani Creek, Sundarbans, West Bengal: A 3.75 MW demonstration project was approved but faced delays due to cost and ecological risks, despite detailed plans.

Wind Energy

Wind energy

- Kinetic energy associated with movement of large masses of air.
- Derives energy from Sun
- Clean , cheap & ecofriendly renewable energy source.
- Harnessed as mechanical energy with the help of wind turbines and is used to operate farm appliances , water pumping & electric power generation.
- Typical speed ranges from 5m/s to 25m/s.

LOCAL WINDS

Produced due to 2 mechanisms:

- Differential heating of land surfaces and water bodies due to solar radiation.
- Differential heating of slopes on the hill sides and that of low lands.

- Wind energy is created when the atmosphere is heated unevenly by the Sun, some patches of air become warmer than others. These warm patches of air rise, other air rushes in to replace them –thus, wind blows.

- A wind turbine extracts energy from moving air by slowing the wind down, and transferring this energy into a spinning shaft , which usually turns a generator to produce electricity.

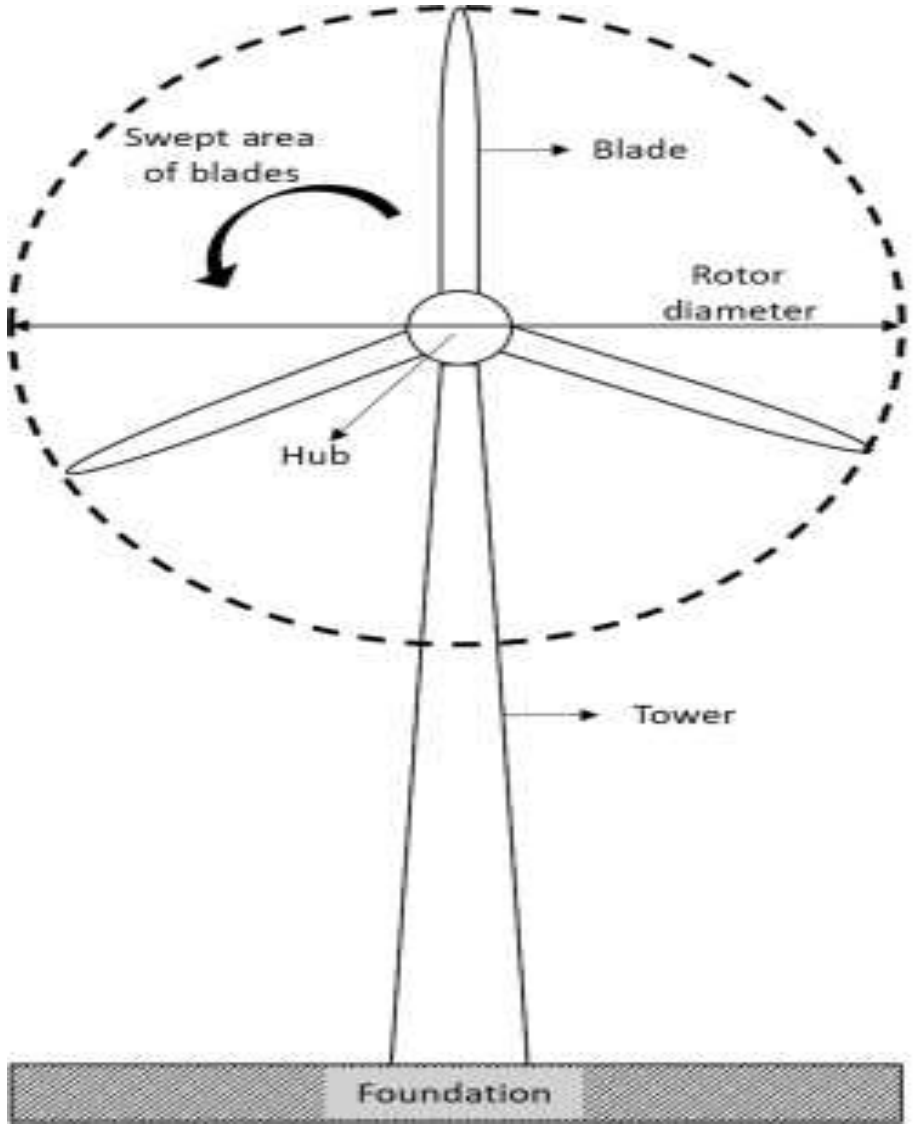
Wind Power

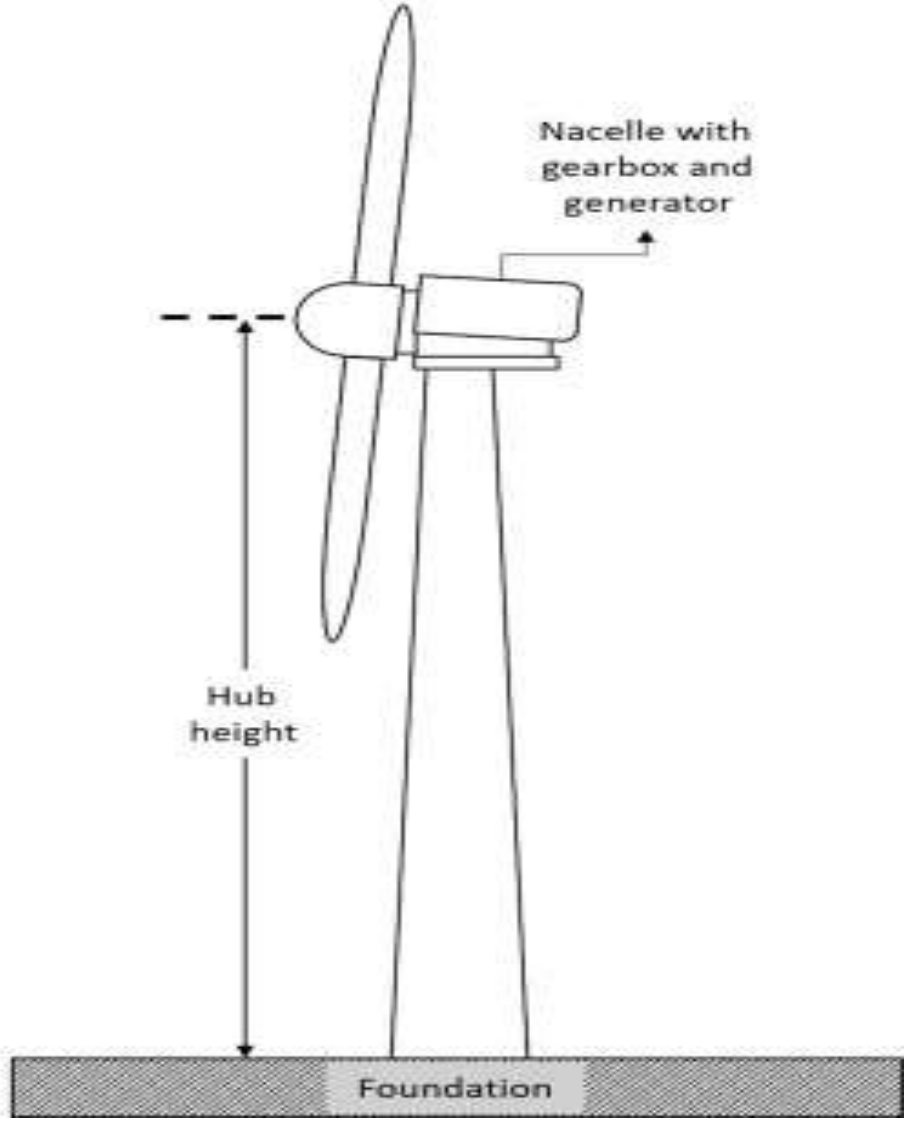
How Does it Work?

Wind power is generated by the force wind exerts on the blades of a turbine, causing the turbine's shaft to rotate at a speed of 10 to 20 revolutions per minute (rpm). The rotor shaft is connected to a generator that converts mechanical energy into electrical energy.

The amount of energy generated by a wind turbine depends on:

- **wind speed** (main factor)
- the **area swept** by the blades
- air **density**


Swept area
of blades
Blade
Rotor
diameter
Hub
Tower
Foundation


Nacelle with
gearbox and
generator
Hub
height
Foundation

Wind turbines require:

- a minimum wind speed (generally 12-14 km/h) to begin turning and generate electricity
- strong winds (50-60 km/h) to generate at full capacity
- winds of less than 90 km/h; beyond that speed, the turbines must be stopped to avoid damage



Since wind speed increases with elevation above ground, wind turbines are installed on towers that are often as tall as a 20-storey building.

- Under optimal conditions, the efficiency of a wind generator in converting energy to electricity is about 45%

History

- Wind energy has been harnessed for centuries to propel sailing vessels and turn grist mills and water pumps.
- Today, wind is used increasingly to generate electricity.
- Turbines with large propellers are erected on ‘wind farms’ located in strategic areas that have good wind regimes and that are in proximity to existing electrical grids.
- Wind energy is captured only when the wind speed is sufficient to move the turbine blades, but not in high winds when the turbine might be damaged if operated

Technology transition “windmills” to “wind turbines”







Basic principles of Wind Energy Conversion Systems (WECS)

Wind possesses energy by virtue of its motion. Any device capable of slowing down the mass of moving air, like a sail or propeller, can extract part of the energy and convert into useful work.

There are three factors determine the output power generated from the wind mill , they are

(1)The wind speed

(2)The cross section of wind swept by rotor, and

(3)The overall conversion efficiency of rotor, transmission system and generator pump.

Components of Wind Energy System

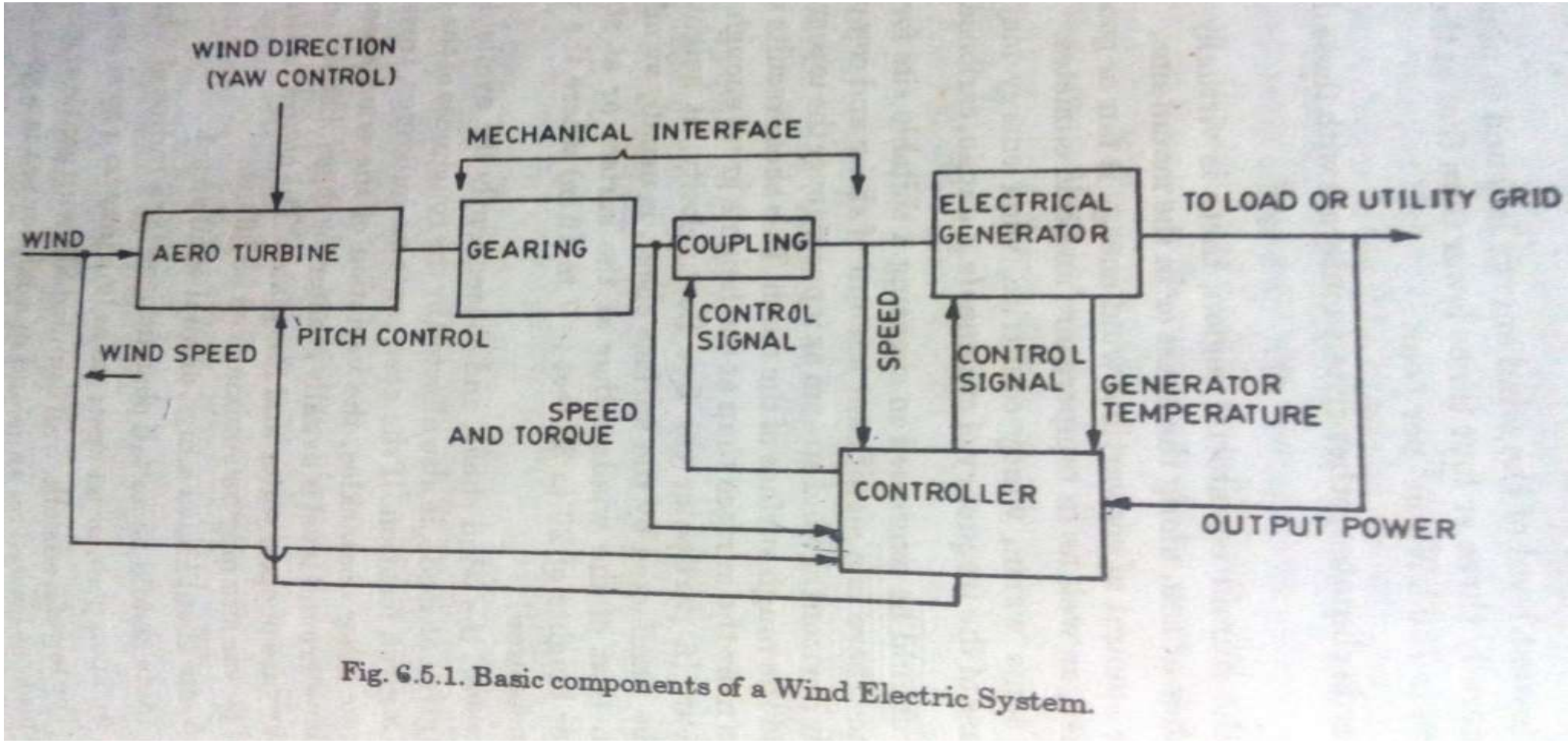
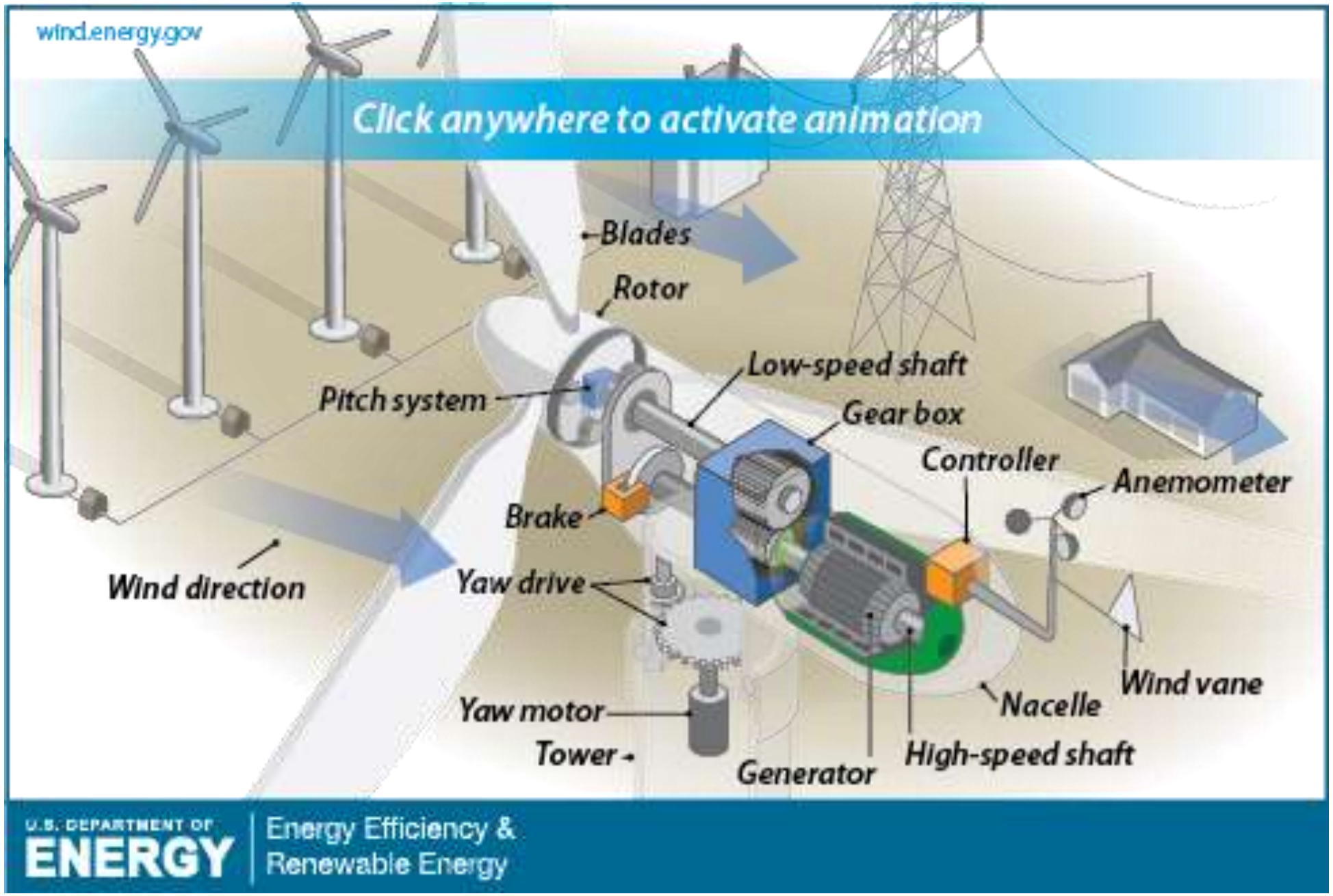


Fig. 6.5.1. Basic components of a Wind Electric System.


wind.energy.gov
Click anywhere to activate animation
Blades
Rotor
Low-speed shaft
Pitch system
Gear box
Controller
Anemometer
Brake
Wind direction
Yaw drive
Wind vane
Nacelle
Yaw motor
Tower
Generator
High-speed shaft
U.S. DEPARTMENT OF ENERGY
Energy Efficiency & Renewable Energy

Anemometer:
Measures the wind speed and transmits wind speed data to the controller.
Blades:
Lifts and rotates when wind is blown over them, causing the rotor to spin. Most turbines have either two or three blades.
Brake: Stops the rotor mechanically, electrically, or hydraulically, in emergencies.
Controller: Starts up the machine at wind speeds of about 8 to 16 miles per hour (mph) and shuts off the machine at about 55 mph. Turbines do not operate at wind speeds above about 55 mph because they may be damaged by the high winds.

Gear box:

Connects the low-speed shaft to the high-speed shaft and increases the rotational speeds from about 30-60 rotations per minute (rpm), to about 1,000-1,800 rpm; this is the rotational speed required by most generators to produce electricity.

Generator:

Produces 50-cycle AC electricity; it is usually an induction generator.

High-speed shaft: Drives the generator.

Low-speed shaft: Turns the low-speed shaft at about 30-60 rpm.

Nacelle: Sits atop the tower and contains the gear box, low- and high-speed shafts, generator, controller, and brake.

Pitch: Turns (or pitches) blades out of the wind to control the rotor speed, and to keep the rotor from turning in winds that are too high or too low to produce electricity.

Rotor: Blades and hub together form the rotor.

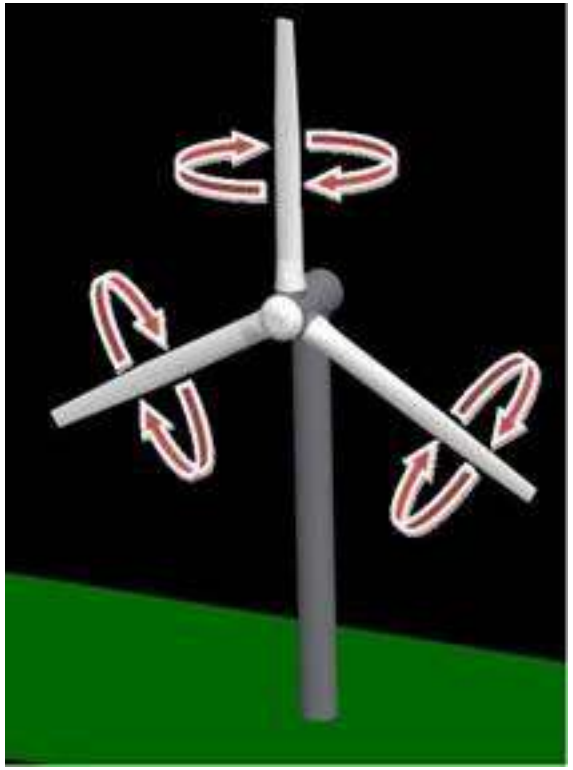
Tower: Made from tubular steel (shown here), concrete, or steel lattice. Supports the structure of the turbine. Because wind speed increases with height, taller towers enable turbines to capture more energy and generate more electricity.

Wind vane: Measures wind direction and communicates with the yaw drive to orient the turbine properly with respect to the wind.

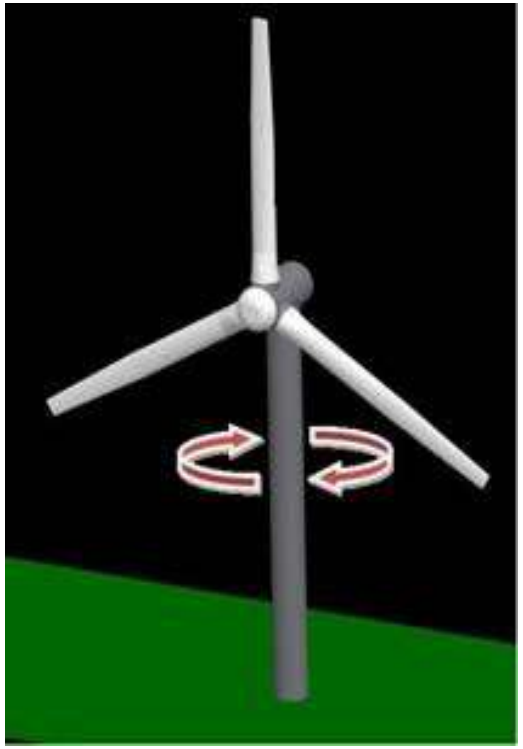
Yaw drive: Orients upwind turbines to keep them facing the wind when the direction changes. Downwind turbines don't require a yaw drive because the wind manually blows the rotor away from it.

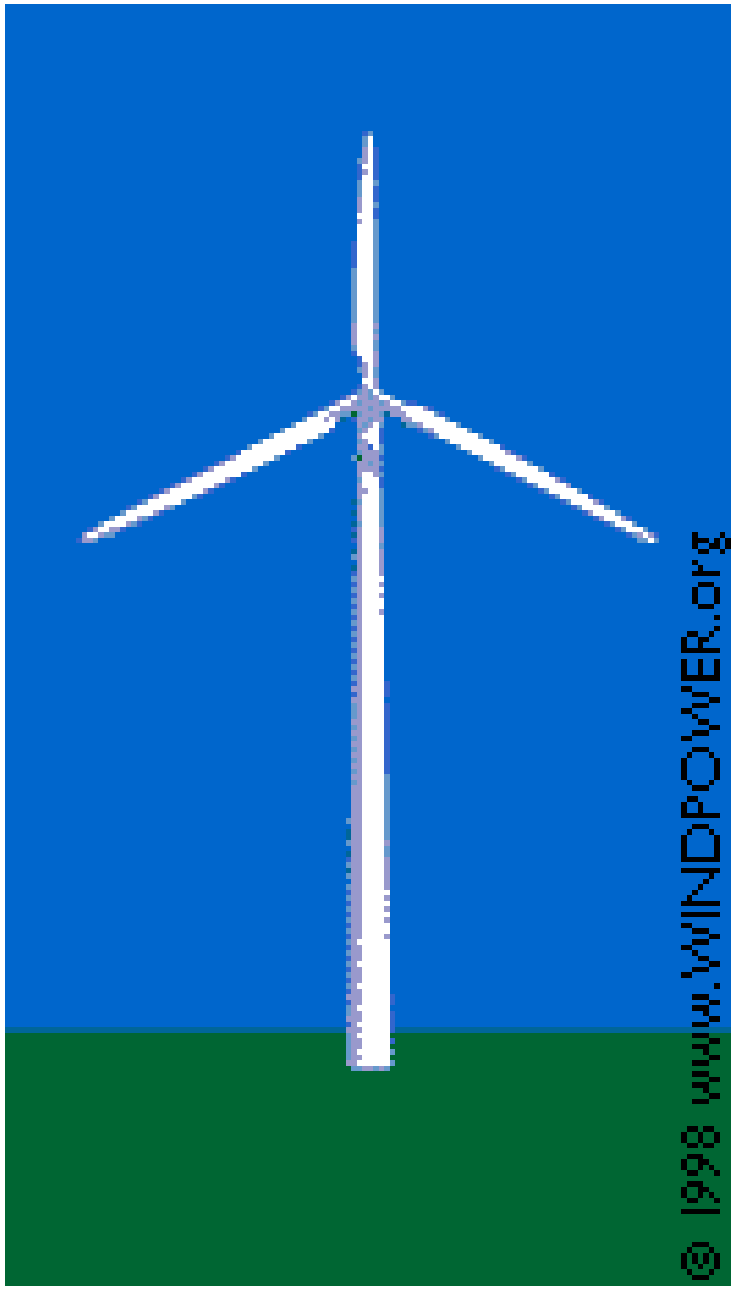
Yaw motor: Powers the yaw drive.

Pitch control



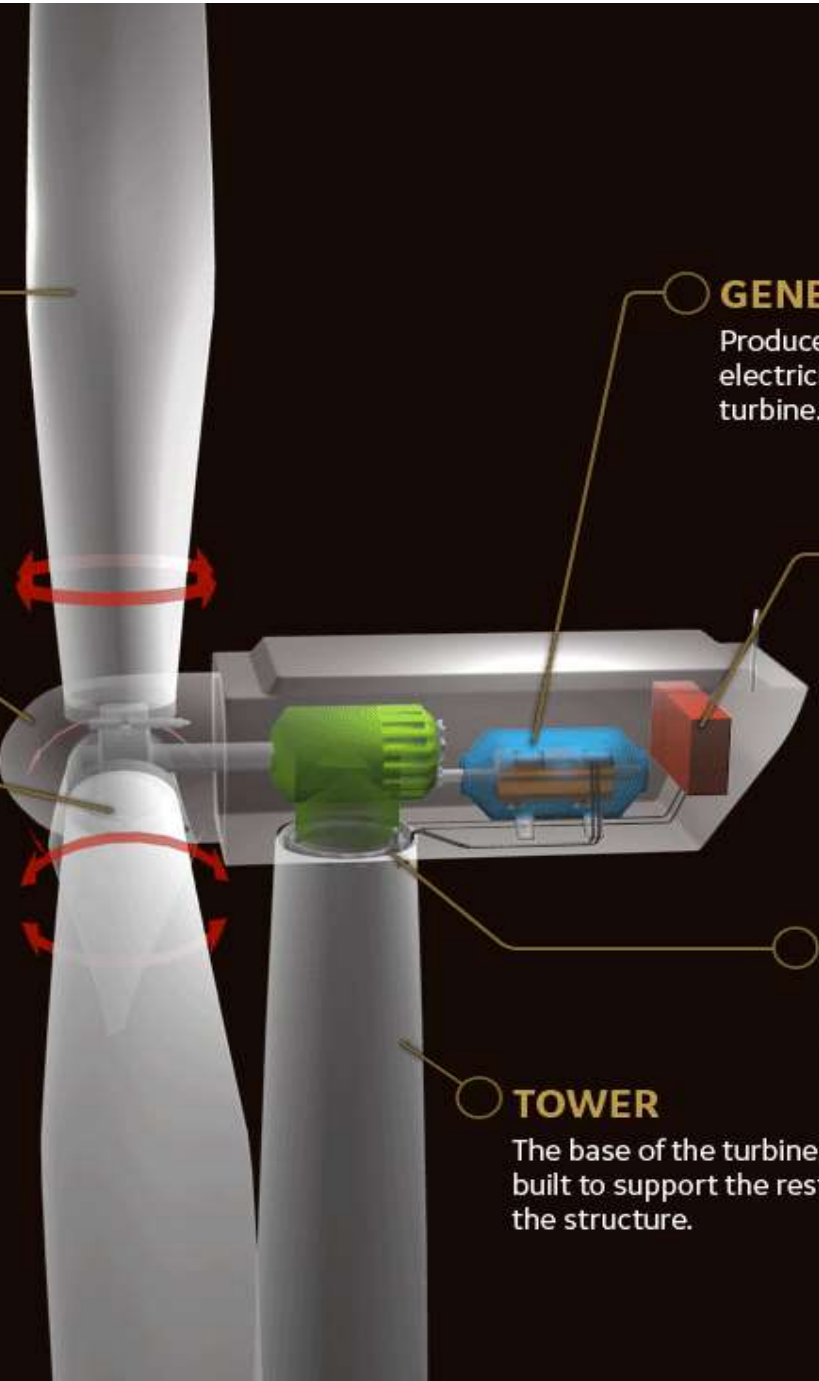
Yaw control





Yaw
Control

GE WIND TURBINE



BLADES
Lift and rotate when hit by wind, causing the rotor to spin.

ROTOR
Combination of the blades and hub.

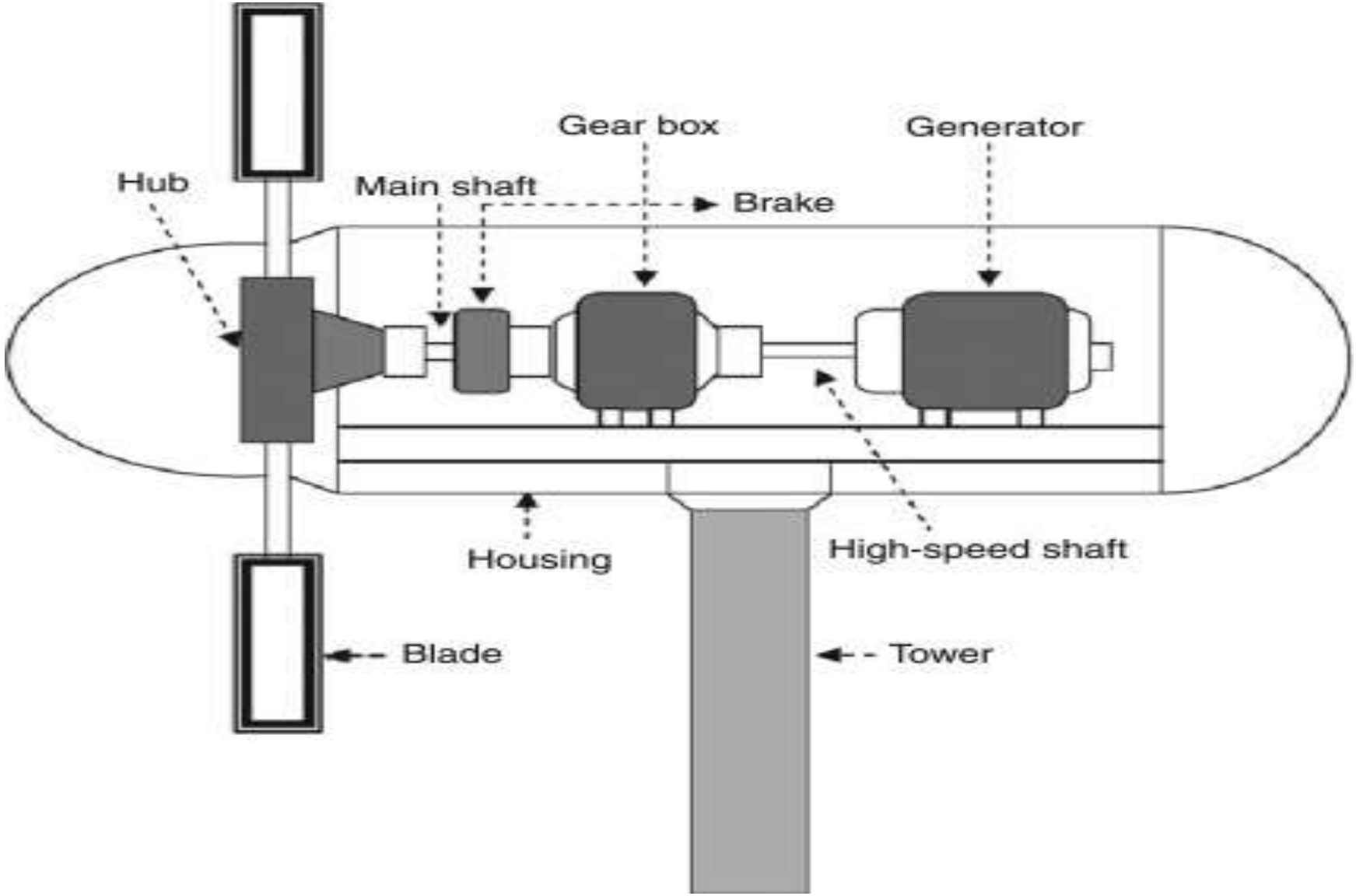
PITCH SYSTEM
Turns blades out of the wind to control rotor speed. Also, stops the rotor from spinning in conditions where wind is blowing too slow or too fast.

GENERATOR
Produces 60-cycle AC electricity within the turbine.

CONTROLLER
Starts and stops the turbine from working, depending on conditions.

YAW DRIVE
Controls upwind turbines to orient them should wind direction change.

TOWER
The base of the turbine, built to support the rest of the structure.


Gear box
Generator
Hub
Main shaft
Brake
Housing
High-speed shaft
Blade
Tower

- The purpose of pitch control is to maintain the optimum blade angle to achieve certain rotor speeds or power output.
- You can use pitch adjustment to stall and furl, two methods of pitch control.
- By stalling a wind turbine, you increase the angle of attack, which causes the flat side of the blade to face further into the wind.
- Furling decreases the angle of attack.
- Pitch angle adjustment is the most effective way to limit output power by changing aerodynamic force on the blade at high wind speeds.
- Yaw refers to the rotation of the entire wind turbine in the horizontal axis. Yaw control ensures that the turbine is constantly facing into the wind to maximize the effective rotor area and, as a result, power.

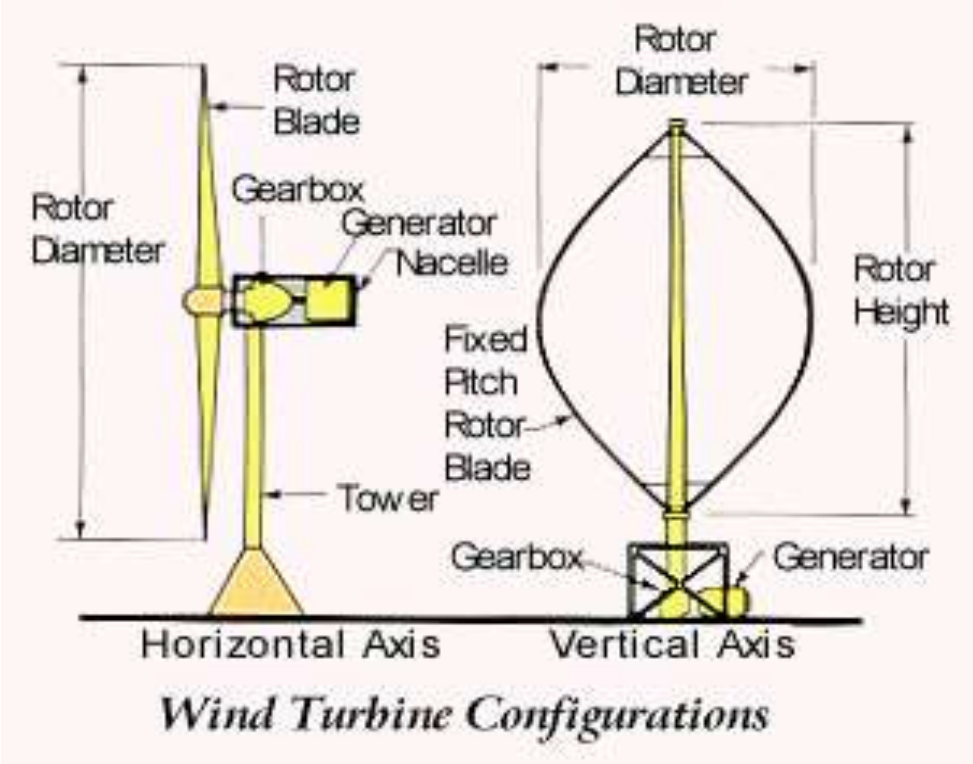
Wind speed fluctuates, which has an impact on wind electricity generation capacity and operating characteristics. In general, wind speeds are as follows:

- 8 kph (2 m/s) minimum is required to start rotating most small wind turbines.
- 12.6 kph (3.5 m/s) is the typical cut-in speed, when a small turbine starts generating power.
- 36–54 kph (10–15 m/s) produces maximum generation power.
- At 90 kph (25 m/s) maximum, the turbine is stopped or braked (cut-out speed).

Classification

Types Based on orientation of the axis of rotation of the rotor

- Horizontal axis wind turbine (HAWT)
- Vertical axis wind turbine (VAWT)



Wind Turbine Configurations

Horizontal axis wind turbine (HAWT)

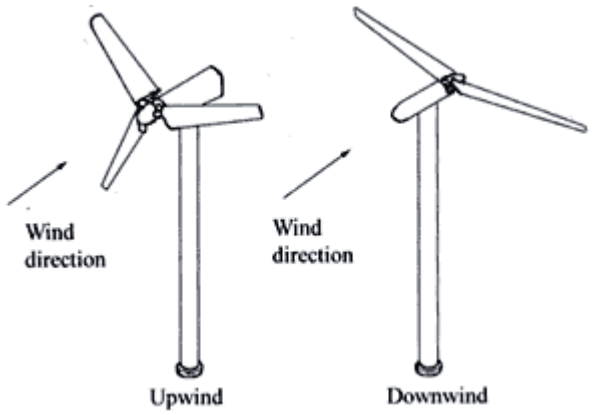
- HAWTs have their shafts mounted horizontally parallel to the ground.
- HAWTs are the most common and usable wind turbine type .
- The electric generator and the rotor shaft, which is fitted horizontally to the ground, are placed on the top of a tall tower.
- The blades of the rotor are forced to rotate due to the air flow.
- The rotor shaft meshes with the generator, and consequently, the rotor shaft rotation leads to electricity generation..

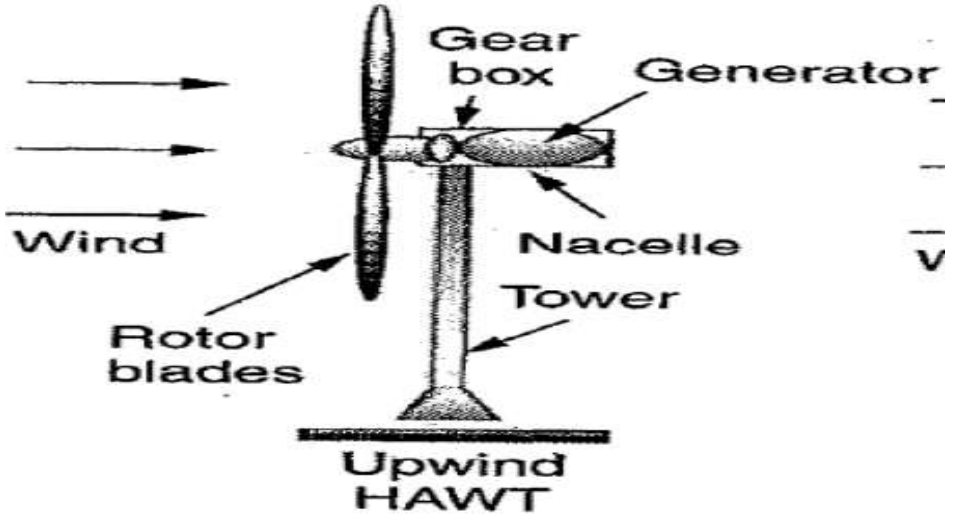
- This type of turbine is usually designed to ensure that the rotor blades are facing the wind , thus it utilizes wind sensor and servo motor.
- High wind speeds could be dangerous and may cause damage to the wind turbine; to prevent that, the turbine is fitted with a brake to reduce rotor shaft speed.
- The main HAWT advantages are the tall towers, which result in more powerful wind with wind shear, leading to an upsurge in the turbine output power.
- Moreover, they can be installed offshore, in forests above trees or uneven land.
- The disadvantages of the HAWTs is that they need a brake mechanism to slow down the rotor blades in the case of strong wind and a supplementary mechanism is required to allow the rotor blades to be turned in the direction of the wind

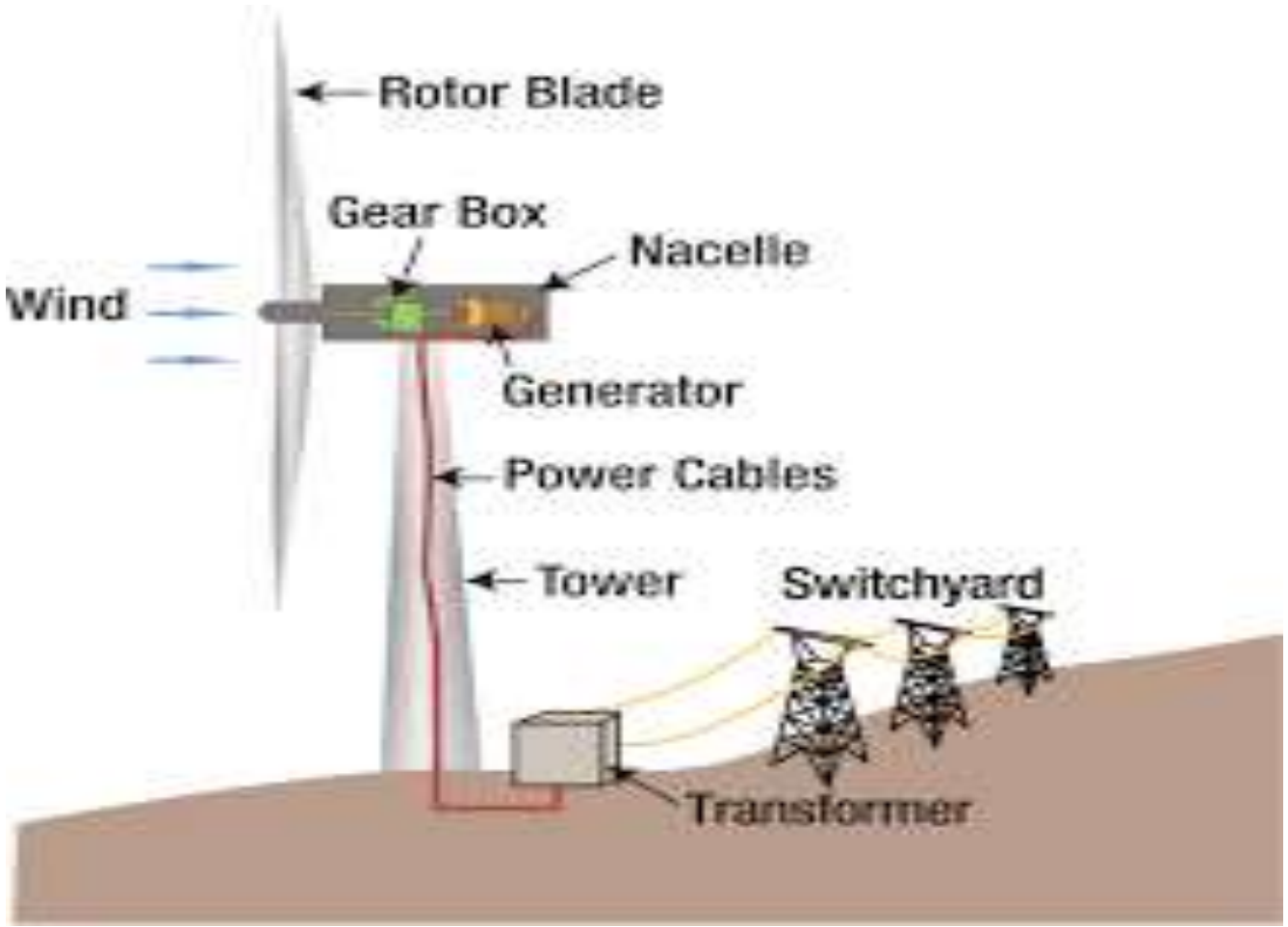
- Windmills have multiple blades
  - need to provide high starting torque to overcome weight of the pumping rod
  - must be able to operate at low windspeeds to provide nearly continuous water pumping
  - a larger area of the rotor faces the wind
- Turbines with many blades operate at much lower rotational speeds - as the speed increases, the turbulence caused by one blade impacts the other blades
- Most modern wind turbines have two or three blades

Horizontal Axis Wind Turbines

- "Upwind" HAWT –blades are in front of (upwind of) the tower
- Most modern wind turbines are of this type
- Blades are "upwind" of the tower
- Require somewhat complex yaw control to keep them facing into the wind
- Operate more smoothly and deliver more power






Rotor Blade
Gear Box
Nacelle
Wind
Generator
Power Cables
Tower
Switchyard
Transformer

Types of Horizontal axis turbine







Horizontal axis wind turbines (HAWT) classified based on the number of blades



1.Single-Blade Turbines

- A long blade is mounted on a rigid hub
- If extremely large blades are mounted on rigid hub, large blade root bending moments may occur due to tower shadow, gravity and sudden shifts in wind directions.
- To reduce rotor cost ,low cost counter weight are recommended.

Advantages

- 1. Least expensive
- 2.Simple blade controls
- 3. Easiest to erect because of its lightweight and because the blade can be mounted while it is on the ground
- 4. Requires the smallest and lightest tower

Disadvantages

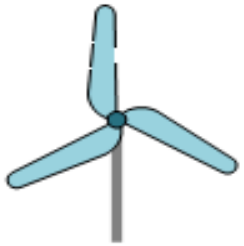
- 1. Noisier than the three-blade turbine
- 2. Must run at the highest speed to produce the same amount of electrical power
- 3. Vibration produced at the blade
- 4.Large blade root bending moment
- 5.Unconventional appearance



Single
Blade



Two Blade



Three Blade

Two-Blade Turbine

- Rotor drives generator through a step up gearbox
- Two-blade turbines have experienced high fatigue failures of the blade and other mechanical parts, so they have limited application

Advantages

1. Initial cost and weight are lower and they are simpler to mount
2. Produces more energy than the single-blade turbine

Disadvantages

1. Noisier than the three-blade turbine
2. Produces less energy than the three-blade turbine(when blade size and speed are the same)

- 17/12/25 4th
- 13,18,21,25,62,64,69

Three-Blade Turbine

•majority of large horizontal-axis wind turbines use three blades, with the rotor position maintained upwind by the yaw control.

•provides the most energy conversion while limiting noise and vibration.

•The three blades provide more blade surface for converting wind energy into electrical energy than a two-blade or single-blade wind turbine.

Advantages

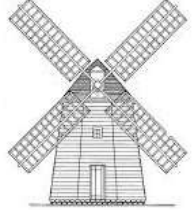
1. Quietest of the three types of turbines
2. Least amount of vibration
3. Available blade pitch control allows the blade to catch the maximum amount of wind
4. Lowest energy cost

Disadvantages

1. Heavier than single-and two-blade turbines
2. Most capital-expensive of the three types
3. Requires active yaw control to make blades face into the wind
4. Requires the largest cranes to construct
5. Requires the largest and heaviest tower
6. Larger blades are more difficult to transport to the tower site

- Single-bladed designs are not very popular due to problems in balancing and visual acceptability
- Two-bladed rotors also have these drawbacks, but to a lesser extent.
- Most of the modern wind turbines employed for electricity generation have three-bladed rotors.
- The loading pattern for these rotors is relatively uniform and they are visually more acceptable.

Horizontal axis Dutch type Windmill



•One of the oldest designs

•Blades are made from an array of wooden slats which feather at high wind speeds

Horizontal axis Multiblade Windmill:

•Modern water pumping windmills have a large number of blades-generally wooden or metallic-driving a reciprocating pumps. (water pumping)

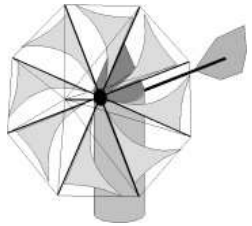
•Have high strength to weight ratio

•Have good power coefficient

•High starting torque

•Simplicity and low cost





Sail type

•is of Recent origin

•Blade surface is made from cloth, nylonor plastics arranged sail wings

Multi blade rotor

- Fabricated from curved sheet metal blades
- Width of blades increases outwards from the centre

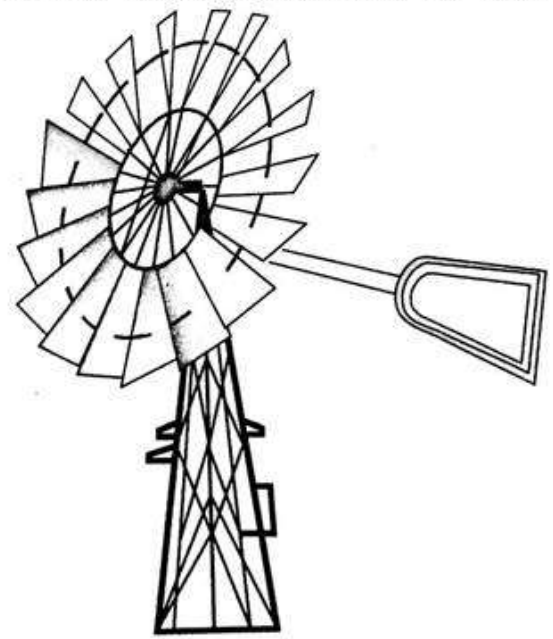
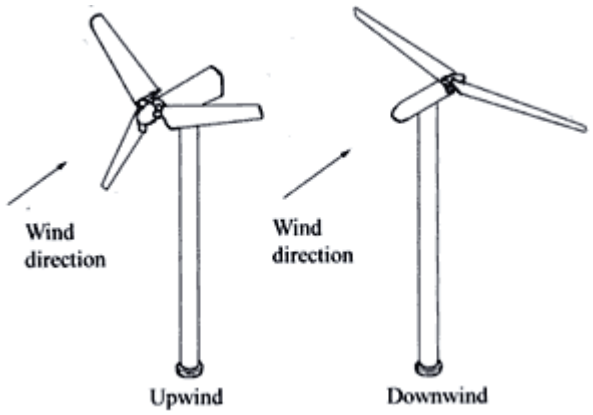


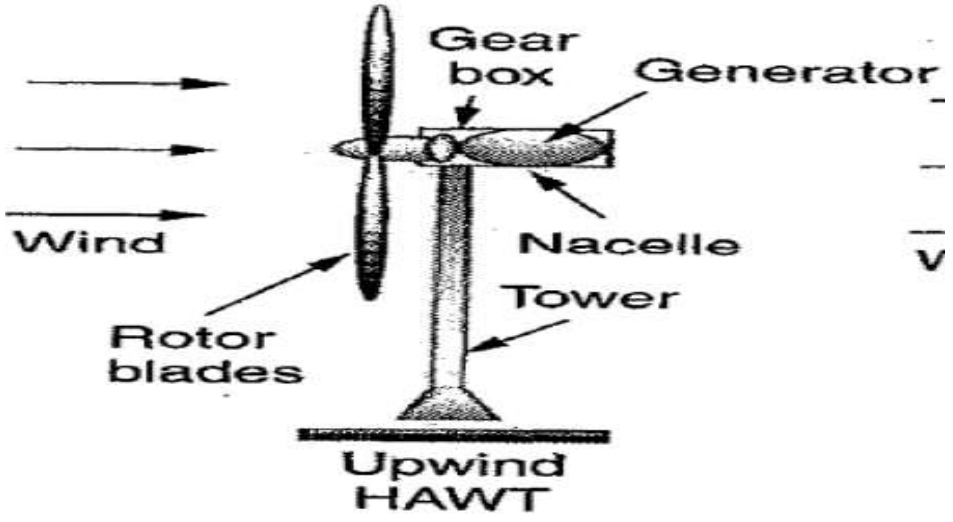
Figure 7.2 Multiblade rotor installed on a tower.

- Further, a HAWT can have upwind- or downwind-type rotors.
- An upwind turbine has its rotor fixed in front of the unit, directly facing the incoming wind stream .
- In contrast, the downwind turbines have their rotors positioned at the back side, leaving the nacelle to face the wind first. The major advantage of upwind rotors is that they do not suffer from the tower shadow effect.
- However, upwind rotors are to be placed at some distance from the tower and a yaw mechanism is essential to keep the rotor always facing the wind.
- On the other hand, downwind machines are more flexible and may not require the yaw mechanism.
- This makes these designs relatively cheaper. But, as the rotors are placed at the leeward side of the tower , there may be uneven loading on the blades as they pass through the shadow of the tower.

Horizontal Axis Wind Turbines

- "Upwind" HAWT –blades are in front of (upwind of) the tower
- Most modern wind turbines are of this type
- Blades are "upwind" of the tower
- Require somewhat complex yaw control to keep them facing into the wind
- Operate more smoothly and deliver more power





Advantages of Horizontal Axis Machines

- The efficiency is higher than that of vertical axis machines since the blades always move perpendicularly to the wind, receiving power through the whole rotation.
- The tall tower base allows access to stronger wind in sites with wind shear
- They are easier to mount high
- They are self starting.
- They are less expensive.
- The technology is better developed.
- They are available commercially.

Disadvantages of Horizontal Axis Machines

- Massive tower construction is required to support the heavy blades, gearbox, and generator.
- Many of the important parts that require maintenance are high off the ground.
- A yaw mechanism must be in place to turn the turbine into the wind.
- HAWTs generally require a braking or yawing device in high winds to stop the turbine from spinning and destroying or damaging itself.
- Cyclic Stresses & Vibration

Terms used in wind energy

- **TSR** refers to the ratio between the **wind** speed and the speed of the tips of the **wind turbine** blades.
- **Wind turbines** must be designed with optimal tip speed ratios to get the maximum amount of power from the **wind**.
- TSR (λ) =Tip Speed of Blade/ Wind Speed

- Drag Force

Force component in line with velocity of the wind.

- Lift Force

Force component perpendicular to drag force.

- Cut in Speed

It is the wind speed at which a wind turbine starts to operate.

- Cut Out Speed

It is the wind speed at which a wind turbine is designed to shut down to prevent damage from high wind speeds. It is also known as furling wind speed.

- Blade designs operate on either the principle of drag or lift.
- For the drag design, the wind literally pushes the blades out of the way. Drag powered wind turbines are characterized by slower rotational speeds and high torque capabilities. They are useful for the pumping, sawing or grinding work.
- The lift blade design employs the same principle that enables airplanes, kites and birds to fly. The blade is essentially an airfoil, or wing. When air flows past the blade, a windspeed and pressure differential is created between the upper and lower blade surfaces.
- The pressure at the lower surface is greater and thus acts to "lift" the blade. When blades are attached to a central axis, like a wind turbine rotor, the lift is translated into rotational motion. Lift-powered wind turbines have much higher rotational speeds than drag types and therefore well suited for electricity generation.

- The tip-speed is the ratio of the rotational speed of the blade to the wind speed.
- The larger this ratio, the faster the rotation of the wind turbine rotor at a given wind speed.
- Electricity generation requires high rotational speeds.
- Lift-type wind turbines have maximum tip-speed ratios of around 10, while drag-type ratios are approximately 1.
- Given the high rotational speed requirements of electrical generators, it is clear that the lift-type wind turbine is most practical for this application.

- The rated speed is the minimum wind speed at which the wind turbine will generate its designated rated power.
- For example, a "10 kilowatt" wind turbine may not generate 10kilowatts until wind speeds reach 40 kmph. Rated speed for most machines is in the range of 40 to 55 kmph.
- At wind speeds between cut-in and rated, the power output from awind turbine increases as the wind increases.
- Most manufacturers provide graphs, called "power curves,"showing how their wind turbine output varies with wind speed.

Maximum Power Coefficient:-

- *The maximum power coefficient of the wind energy can be defined as the ratio of the convertible power to the theoretically maximum power from the available wind energy.*
- *In practice power coefficients of 20-30% are common*

Betz limit

- It is the flow of air over the blades and through the rotor area that makes a wind turbine function. The wind turbine extracts energy by slowing the wind down.
- The theoretical maximum amount of energy in the wind that can be collected by a wind turbine's rotor is approximately 59%. This value is known as the Betz limit.

Vertical Axis Wind Turbines

- Rotor axis is vertical and remains perpendicular to the wind stream
- Darrieus rotor -the only vertical axis machine with any commercial success
- The main advantage of this arrangement is that the wind turbine does not need to be pointed into the wind. This is an advantage on sites where the wind direction is highly variable or has turbulent winds.
- No yaw (rotation about vertical axis) control needed to keep them facing into the wind
- Heavy machinery in the nacelle is located on the ground
- Blades are closer to ground where wind-speeds are lower

Vertical Axis Wind Turbine (VAWT)

- VAWTs are in the development stage and many models are undergoing field trial.
- Main attractions of a VAWT are:
  1. it can accept wind from any direction, eliminating the need of yaw control.
  2. gearbox, generator etc. are located at the ground, thus eliminating the heavy nacelle at the top of the tower. This simplifies the design and installation of the whole structure, including tower.
  3. the inspection and maintenance also gets easier and it also reduces the overall cost.

1. Main Components

- The constructional details of a vertical axis wind turbine (Darrieus type rotor) are shown in Fig. 7.23.
- The details of main components are as follows:
  a) Tower (or Rotor Shaft)
  b) Blades
  c) Support Structure

Vertical Axis Wind Turbine (VAWT) (Cont..)

a) Tower (or Rotor Shaft): The tower is a hollow vertical rotor shaft, which rotates freely about vertical axis between top and bottom bearings. It is installed above a support structure. In the absence of any load at the top, a very strong tower is not required, which greatly simplifies its design. The upper part of the tower is supported by guy ropes. The height of the tower of a large turbine is around 100 m.

b) Blades It has two or three thin, curved blades shaped like an eggbeater in profile, with blades curved in a form that minimizes the bending stress caused by centrifugal forces-the so-called 'Troposkien' profile. The blades have air foil cross-section with constant chord length. The pitch of the blades cannot be changed. The diameter of the rotor is slightly less than the tower height. The first large (3.8 MW), Darrieus type, Canadian machine has rotor height as 94 m and diameter as 65 m with a chord of 2.4 m

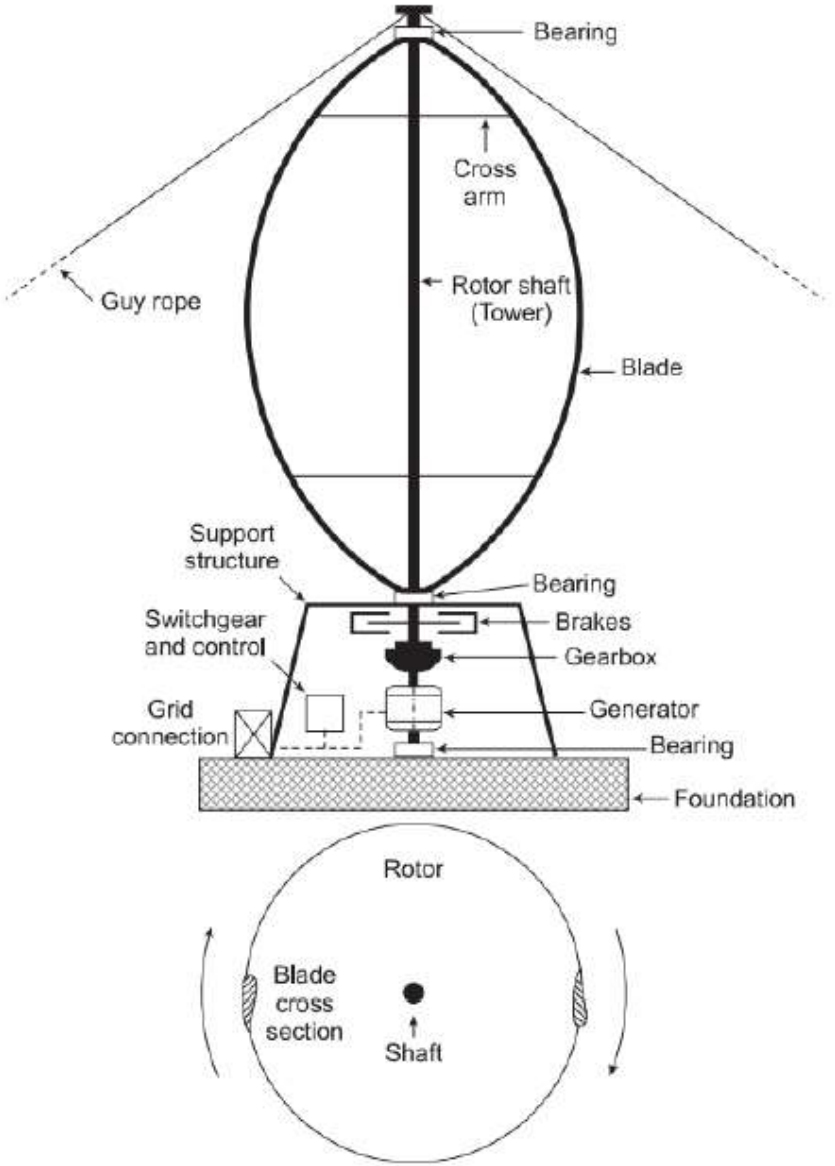


Figure 7.23 Vertical axis wind (Darrieus) turbine

Vertical Axis Wind Turbine (VAWT) (Cont..)

c) Support Structure: Support structure is provided at the ground to support the weight of the rotor. Gearbox, generator, brakes, electrical switchgear and controls are housed within this structure.

2. Types of Rotors

- Various types or rotors for VAWTs are shown in Fig. 7.24.
- The simplest being **three or four cups structure** attached symmetrically to a vertical shaft. Drag force on concave surface of the cup facing the wind is more than that on convex surface. As a result, the structure starts rotating. Some lift force also helps rotation. However, it cannot carry a load and is, therefore, not used as power source. Main characteristic of this rotor is that its rotational frequency is linearly related to wind speed. Therefore, it is used as a transducer for measuring the wind speed and the apparatus is known as **cup anemometer**.
- The **Savonius or S-rotor** consists of two half cylinders attached to a vertical axis and facing in opposite directions to form a two-vaned rotor. It has high starting torque, low speed and low efficiency. It can extract power even from very slow wind, making it working most of the time. These are used for low power applications. High starting torque particularly makes it suitable for pumping applications, using positive displacement pumps.

Vertical Axis Wind Turbine (VAWT) (Cont..)

- Darrieus rotor is used for large-scale power generation. Power coefficient is considerably better than S-rotor. It runs at a large tip-speed ratio. The aerodynamic force on the blade reverses in every revolution causing fatigue. This along with centrifugal force complicates the design of the blade. One of the drawbacks of this rotor is that it is usually not self-starting. Movement may be initiated by using electrical generator as motor. As the pitch of the blade cannot change, the rotor frequency and thus the output power cannot be controlled. Rotor frequency increases with wind speed and power output keeps on increasing till the blades stall. Hence at high wind speed it becomes difficult to control the output. For better performance and safety of the blades, gearbox and generator, etc., it is desirable to limit the output to a level much below its maximum possible value.

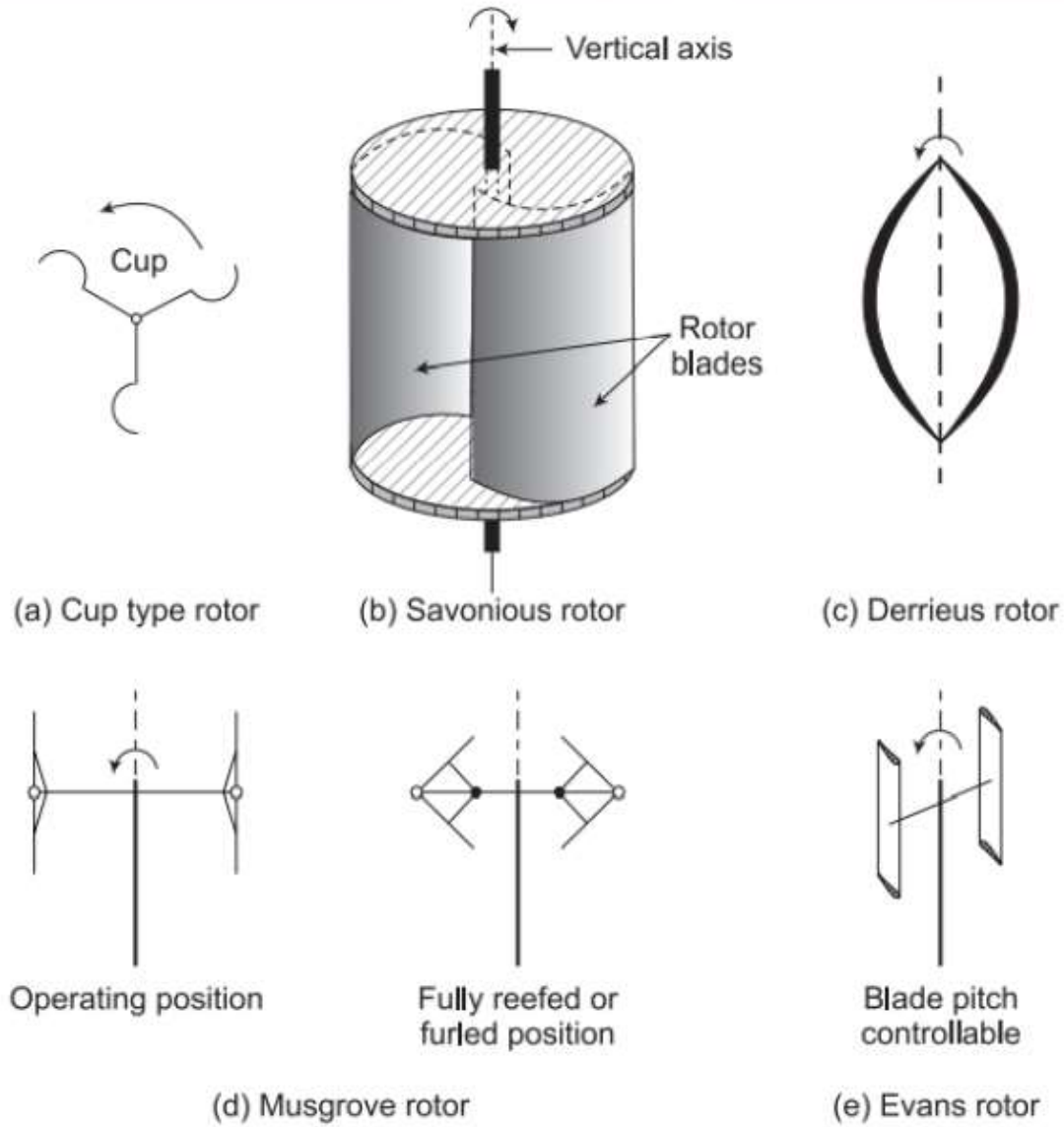


Figure 7.24 Various types or rotors for VAWT

Disadvantages of Vertical Axis Machines

- Efficiency is very low compared to HAWTS.

•Most vertical axis machines are not self starting .They need an initial push to start; this initial push that to make the blades start spinning on their own must be started by a small motor.

•They have relative high vibration because the air flow near the ground creates turbulent flow.

•Because of vibration, bearing wear increases which results in the increase of maintenance costs.

•They can create noise pollution.

•VAWTs may need guy wires to hold it up (guy wires are impractical and heavy in farm areas).

•Maintenance is usually more difficult.

Advantages of Vertical Axis Machines

- They can produce electricity in any wind direction.
- Strong supporting tower in not needed because generator, gearbox and other components are placed on the ground.
- Low production cost as compared to horizontal axis wind turbines.
- As there is no need of pointing turbine in wind direction to be efficient so yaw drive and pitch mechanism is not needed.
- Easy installation as compared to other wind turbine.
- Easy to transport from one place to other.
- Low maintenance costs.
- They can be installed in urban areas.
- Low risk for human and birds because blades moves at relatively low speeds.
- They are particularly suitable for areas with extreme weather conditions, like in the mountains where they can supply electricity to mountain huts.





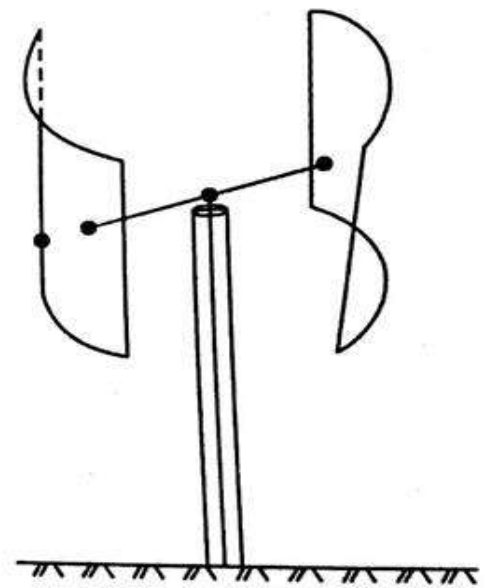


Major VAWT designs

Darrius, Straight-bladed Darrius, Savonious

Savonious rotor

- Comprises two identical hollow semi cylinders fixed to a vertical axis
- Inner side of two half-cylinders face each other to have an S shaped cross section
- Used for water pumping



Darrieus rotor

- 2 or 3 thin curved blades of flexible metal strips
- Operates with wind coming from any direction
- Can be installed close to ground eliminating the cost of the tower structure

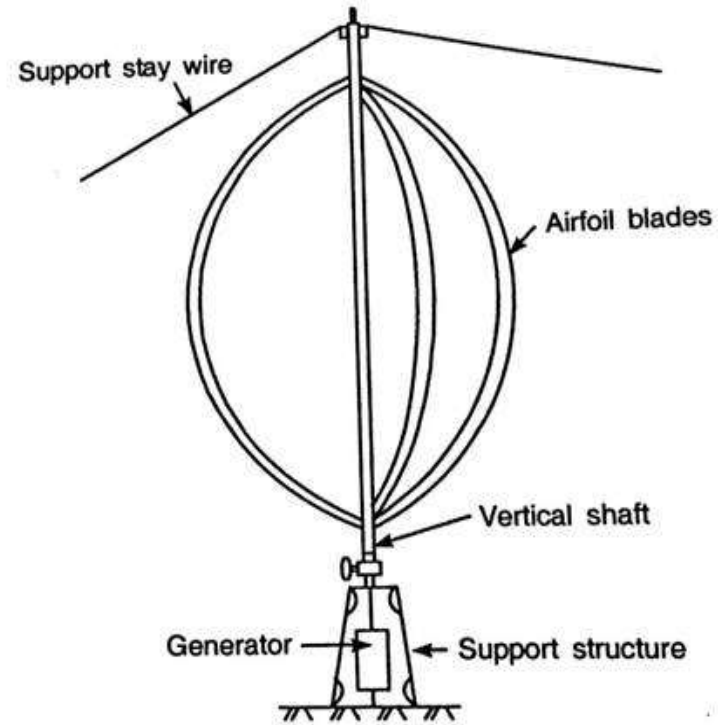
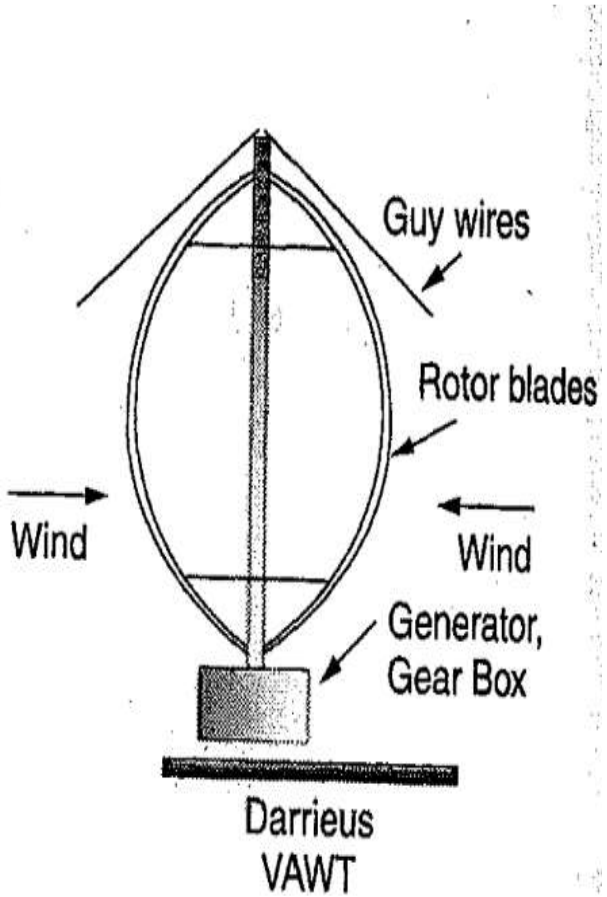


Figure 7.5 Darrieus rotor.


Guy wires
Rotor blades
Wind
Wind
Generator,
Gear Box
Darrieus
VAWT



- 19/12/25 1st hour
- 1,4,5,10,13,14,18,21,25,33,34,52,64,67,68,69,70,71,72

Output power Equation

- Kinetic Energy = 0.5 x Mass x Velocity^2 , where the mass is measured in kg, the velocity in m/s, and the energy is given in joules.
- Air has a known density (around 1.23 kg/m^3 at sea level),

So the mass of air hitting our wind turbine. (which sweeps a known area) each second is given by the following equation: Mass/sec (kg/s) = Velocity (m/s) x Area (m^2) x Density (kg/m^3).

Therefore, the power (i.e. energy per second) in the wind hitting a wind turbine with a certain swept area is given by simply inserting the mass per second calculation into the standard kinetic energy equation given above resulting in the following vital equation:

Power = 0.5 x Swept Area x Air Density x Velocity^3 , where Power is given in Watts (i.e. joules/second), the Swept area in square metres, the Air density in kilograms per cubic metre, and the Velocity in metres per second.

- The equation for wind power(P) is given by

$P = 0.5 \times \rho \times A \times Cp \times V^3 \times Ng \times Nb$

where, ρ = Air density in kg/m3 , A = Rotor swept area (m2). Cp = Coefficient of performance V = wind velocity (m/s) Ng = generator efficiency Nb = gear box bearing efficiency

Wind Turbine Operation and Power versus Wind Speed Characteristics

- The power-speed characteristics of a wind turbine have four separate regions as shown in Fig. 7.25.

a) **Low Speed Region (Zero to Cut-in Speed)** In this region, the turbine is kept in braked position till minimum wind speed (about 5 m/s), known as cut-in speed becomes available. Below this speed the operation of the turbine is not efficient.

b) **Maximum Power Coefficient Region** In this region, rotor speed is varied with wind speed so as to operate it at constant tip-speed ratio, corresponding to maximum power coefficient, CP max. In this range the nature of characteristics is close to that of maximum power available in the wind. The turbine is operated at maximum power output point using pitch control.

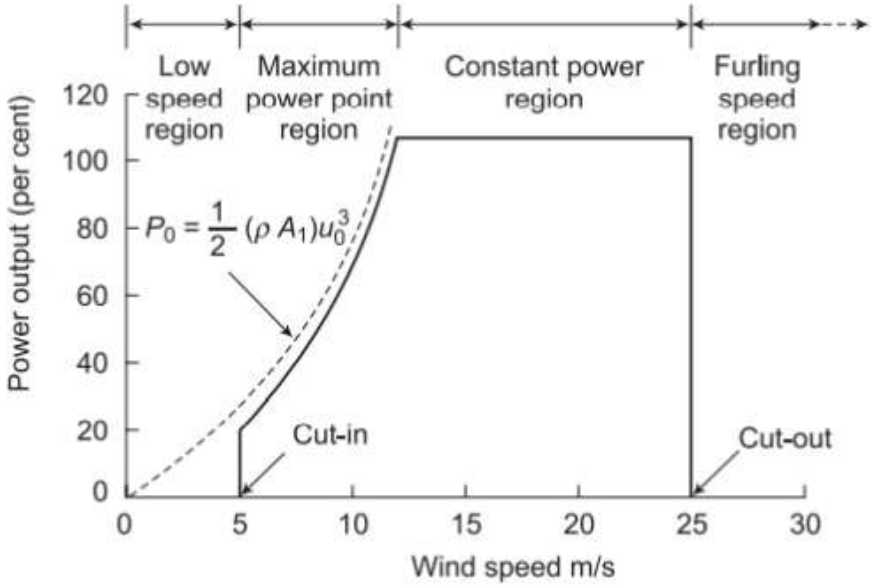


Figure 7.25 Power versus wind speed characteristics

Wind Turbine Operation and Power versus Wind Speed Characteristics (Cont..)

c) **Constant Power Region (Constant Turbine Speed Region)** During highspeed winds (above 12 m/s), the rotor speed is limited to upper permissible value based on the design limits of system components. In this region the power coefficient is lower than CP max. Large machines use pitch control to maintain turbine speed as constant. Such a machine is known as **pitch regulated**. Classical European machines use fixed blades (constant pitch) and the blade twist and thickness are so designed as to crudely maintain the speed as constant. Such a machine is known as stall regulated. This is a simple system requiring only passive technique, as there are no moving blade surfaces or complicated hardware. However, power capture is somewhat less due to rounding of power curve.

- In case of fixed blades rotor, the flow over the blade stalls. The stall performance depends on the twist of blade tip from the root and thickness of blade sections. The lift and drag coefficients change so as to flatten out the peak of the power curve.

Wind Energy Conversion Systems(WECS)

- Converts wind energy into the form of mechanical or electrical energy
- Three modes of operation
  - Standalone mode
  - Back up mode
  - Grid connected mode

Standalone mode

- Isolated from grid
- Feeding power to local loads

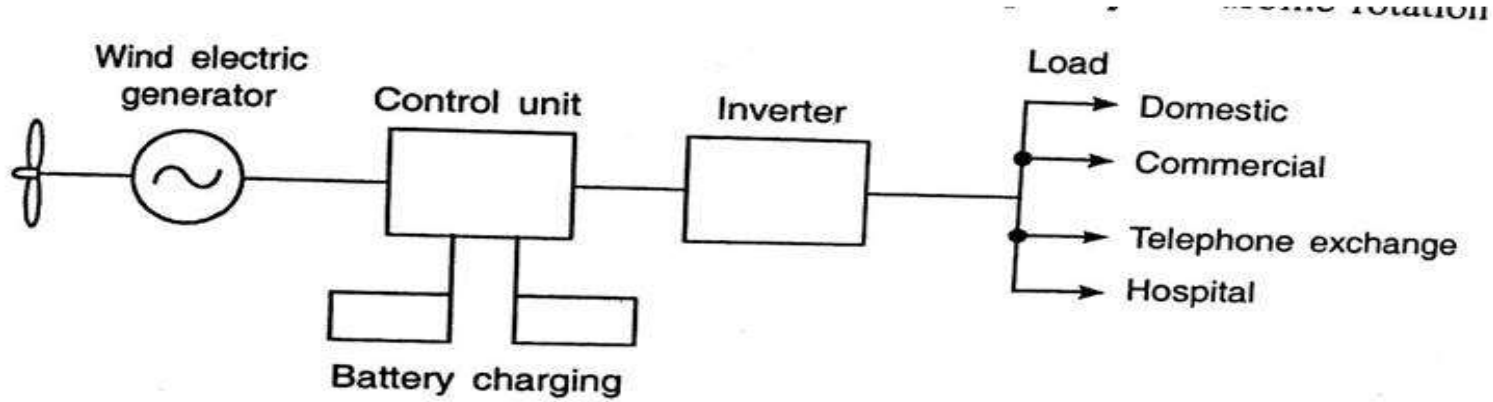


Figure 7.24 A standalone 5 kW wind electric generator.

Back-up mode (wind –diesel)

- Used in areas inaccessible to grid power.
- Combination of wind –diesel power supply
- As wind speed drops low tariff loads are switched off to reduce demand
- During period of no wind, priority loads are fed by DGs.

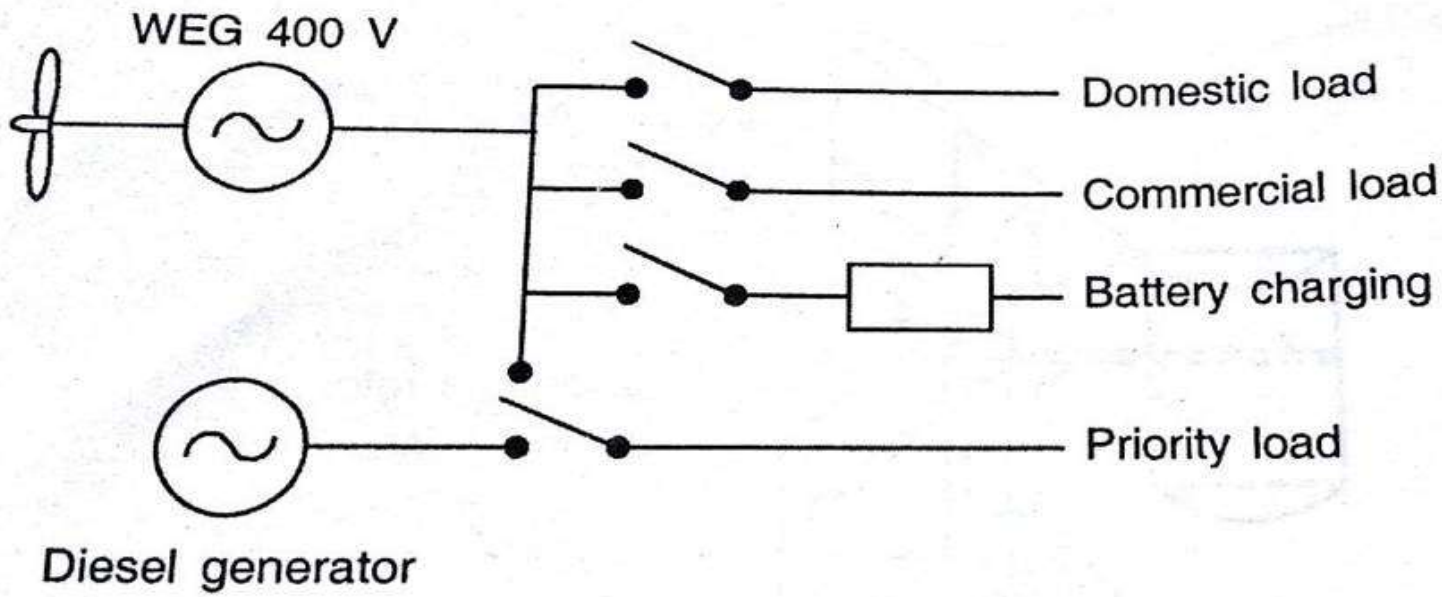


Figure 7.27 Wind–diesel power supply.

Grid connected mode

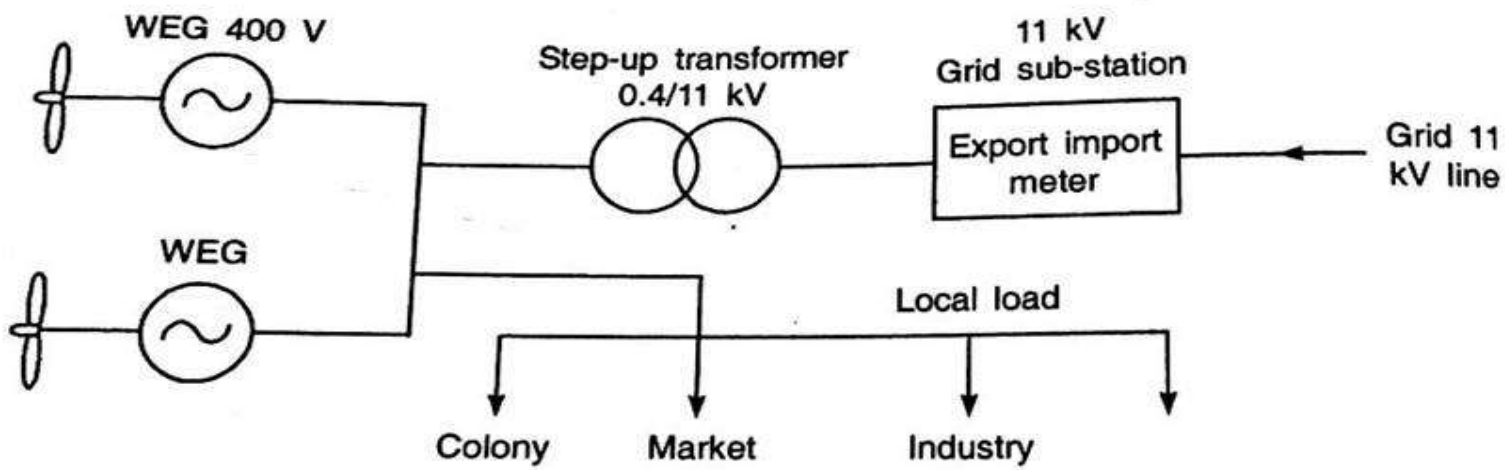


Figure 7.28 Grid-connected wind turbine generators.

Advantages of wind energy

1. It is a renewable source of energy
2. Non polluting-so it has no adverse influence on the environment
3. Wind energy systems avoid fuel provision and transport
4. On a small scale up to a few kilowatt system is less costly On a large-scale costs can be competitive with conventional electricity and lower costs could be achieved by mass production

DISADVANTAGES OF WIND ENERGY

1. Wind energy is fluctuating in nature
2. Unlike water energy wind energy needs storage capacity because of its irregularity.
3. Wind energy systems are noisy in operation ; a large unit *can* be heard many kilometres away.
4. Wind power systems have a relatively high overall *weight* because they involve the construction of a high tower and include a gearbox, a hub and pitch changer, a generator coupling shaft etc.
5. Large areas are needed, typically, propellers 1 to 3m in diameter, deliver power in the 30 to 300 W range.
6. Present systems are neither maintenance free not-practically reliable.

Applications

1. Off grid electrical power
2. Grid connected electrical power
3. Applications requiring mechanical power
   - Heating
   - Sea transport
   - Wind pumps
   - Grinding grain
   - Offshore drilling
   - Navigational signals

BASIC PRINCIPLES OF WIND ENERGY CONVERSION(***very imp)

➢The Nature of the Wind

- The circulation of air in the atmosphere is caused by the non- uniform heating of the earth's surface by the sun.
- Nature of terrain, the degree of cloud cover and the angle of the sun in the sky are all factors which influences this process.

- Average wind speeds are greater in hilly and coastal areas than they are well in land. ie winds also tend to blow more consistently and with greater strength over the surface of the water where there is a less surface drag.
- Wind speeds increase with height. They have traditionally been measured at a standard height of ten metres(20-30% greater),for 60m (30-60% greater)

➢ The power in the Wind

- Wind possesses energy by virtue of its motion
- Three factors determine the output from a wind energy converter :

1. the wind speed
2. the cross-section of wind swept by rotor
3. the overall conversion efficiency of the rotor, transmission system and generator or pump

$$T = \frac{P}{\omega} = \frac{P}{\pi DN}$$

T = torque *kgf* or *Newton (N)*

ω = angular velocity of turbine wheel, m/s

D = diameter of turbine wheel

Capacity Factor

- **The capacity factor is the average power generated, divided by the rated peak power. Let's take a five-megawatt wind turbine**. If it produces power at an average of two megawatts, then its capacity factor is 40% (2÷5 = 0.40, i.e. 40%).

EFFECTS OF WIND SPEED AND GRID CONDITION

Wind generation impacts on the power quality

- Wind energy is an intermittent source of power.
- If a large proportion of the grid load is supplied by wind turbines, the output variations due to wind speed changes can cause voltage variation, flicker effects in normal operation.
- The location and intermittent nature of wind turbine machines can cause power quality problems such as voltage stationary variations (drops, interruptions and rises of voltage (voltage dips)), voltage temporary variations (flicker), harmonics, frequency variations, and low power factor.
- Wind turbines, especially inductive machines, tend to absorb reactive power from the system and produce a low power factor. If wind turbines absorb too much reactive power, the system can become unstable

- Frequency variations are produced when differences in the power balance between generation and demand exist.
- Wind speed variations, connection and disconnection of wind turbines cause differences in the generation; such differences can, in turn, cause frequency variations depending on the power system

Wind Power Equation

$$P = \frac{1}{2}\rho A V^3$$

Where,

P = power,

ρ = air density,

A = swept area of blades given by $A = \pi r^2$

where r is the radius of the blades.

V = velocity of the wind.

A wind turbine travels with the speed is 10 m/s and has a blade length of 20 m. Determine wind power.

Solution:

Given:

Wind speed v =10 m/s,

Blade length l = 20 m,

air density ρ = 1.23 kg/m3,

area , $A = \pi r^2$

$= \pi \times 400$

$= 1256 \ m^2$

The wind power formula is given as,

$$P = \frac{1}{2}\rho A V^3$$

1. It is proposed to design a miniature windmill at a location where wind speed is 11m/s. What is the expected power output in watts for a wind turbine with 6m rotor diameter, coefficient of power is 0.30, η_{gen} =0.80 and that of gearbox is 0.90.

Solution :-

Given –
Diameter of Rotor(D) - 6 m
Coefficient of power (Cp) - 0.3
Generator efficiency (η_{gen}) - 0.8
Gearbox efficiency (η_{gear})- 0.9
Wind speed (u_0) - 11 m/s

$$\text{Actual Power Extracted} = \frac{1}{2}\rho A u_0^3 Cp\, \eta_{gen}\, \eta_{gear}$$

$$= \frac{1}{2} x\, 1.226\, x \frac{\pi}{4} x\, 6^2 x\, 11^3 x\, 0.3\, x\, 0.8\, x\, 0.9$$

$$= 4{,}983\ W$$

2. A 25 kW wind turbine could produce actually only 50,000 kWh per year. Calculate the capacity factor of the system.

Solution :-

Given :

Plant capacity = 25 KW

Actual Production = 50,000 kWh

Now,

Theoretically, the plant should produce

= 25 x 365 x 24 kWh per year

= 2,19,000 kWh / year

$$\text{Capacity Factor (CF)} = \frac{Actual\ Production}{Theoretical\ Production}$$

$$= \frac{50{,}000}{2{,}19{,}000}$$

$$= \mathbf{0.228}$$

therefore, we can say that turbine operates at **22.8%** of its capacity.

1) Ideal Gas Law (Use to find density of wind i.e. air)

$PV = mRT$ or $P = \rho RT$

$$\rho = \frac{P}{RT}$$

(Kg/m^3)

Where, P= Pressure in Pascal, R= Universal Gas Constant = 287 J/Kg K, T= Temperature in K

- 29/12/25
- 3,4,5,7,14,19,25,40,51,57,61,64,69,72

3. A miniature aero-generator is designed to generate an output of 1200 W at wind speed of 5 m/s at one atmospheric pressure and temperature of 20^0C. What will be output if the same aero-generator is installed on the top of the hill where the temperature is 10^0C, pressure is 0.85 atmospheric and wind speed is 6 m/s?

Solution – For case 1, P_{turb} =1200 W

1 atm pressure = 1.01325×10^5

$$\varrho = \frac{P}{RT} = \frac{1.01325 \times 10^5}{287 \times 293} = 1.2049 \text{ kg/m}^3$$

$u_0 = 5$ m/s

Now, power generated by turbine,

$$P_{turb} = \frac{1}{2} \varrho A u_0^3$$

$$1200 = \frac{1}{2} \times 1.2049 \times A \times 5^3$$

$A = 15.935 \text{ m}^2$.....1

For Case 2 , 0.85 atm pressure

$= 0.85 \times 1.01325 \times 10^5$

$$\varrho = \frac{P}{RT} = \frac{0.85 \times 1.01325 \times 10^5}{287 \times 283} = 1.0604 \text{ kg/m}^3$$

$u_0 = 6$ m/s

Power generated by turbine,

$$P_{turb} = \frac{1}{2} \varrho A u_0^3$$

$$= \frac{1}{2} \times 1.0604 \times 15.935 \times 6^3$$

P_{turb}= 1824.92 W

Calculate the rotor radius & rotor speed for multi-blade wind machine operating at a design speed of 25 kmph. The machine operates a water pump having capacity of 5.1 m^3/hr. and lift of 9m. Following data is given: Density of water = 996 kg/m^3, Efficiency of Water Pump =0.6, Efficiency of transmission from rotor to pump = 0.9, Cp = 0.31, Tip Speed Ratio λ =0.75, Density of air = 1.2 kg/m^3

- V=25km/hr= 25x1000/3600=6.94m/s
- Q= 5.1 m3/hr= 5.1/3600= 1.417x10-3 m3/s
- H=9 m
- ρw=996kg/m3
- Poutput pump= ρgQH

=996*9.81*1.417x10-3x9

=124.58W



$$\eta_{Overall} = \eta_{wp}\,\eta_{tr}$$

$$\eta_{Overall} = \frac{Output}{Input} = \frac{P_{Output-pump}}{P\ (Wind\ Turbine)}$$

$$\eta_{wp}\,\eta_{tr} = \frac{P_{Output-pump}}{P\ (Wind\ Turbine)}$$

$$P\ (Wind\ Turbine) = \frac{P_{Output-pump}}{\eta_{wp}\,\eta_{tr}} = \frac{124.58}{0.9\ x\ 0.6}$$

$P = 230.7$ W

(This is the power that wimd turbine needs to generate)

$$C_p = \frac{\text{Power deliverd by turbine rotor, } P}{\text{Total Wind Power, } P_T} \qquad C_p = \frac{P}{P_T}$$

$$0.\dot{3}1 = \frac{230.70}{\frac{1}{2}\rho A_1 u_0^3}$$

(*Therefore* A_1 = **3.7107** m^2

$$A_1 = \frac{\pi}{4} D^2$$

(*Therefore* D = **2.17** m^2

(***Therefore Rad****ius of Rotor* r = **1.087** m

$$\lambda = \frac{\text{Blade Tip Speed, } u_b}{\text{Free Stream wind Velocity, } u_0}$$

Where, $u_b = R\omega$

$$0.75 = \frac{1.087 \times \omega}{6.94} \quad \textit{Therefore}\ \omega = 4.79 \text{ rad/s}, \quad \omega = \frac{2\pi N}{60} \quad N = 45.74 \text{ rpm}$$

SITE SELECTION CONSIDERATIONS

- Wind energy conversion machines should be located preferable in areas where the winds are strong and persistent

➢ *High annual average wind speed*

- The wind velocity is the critical parameter
- $PW = KV^3$

➢ *Availability of anemometry data*

- The principal object is to measure the wind speed which basically determines the WECS output power
- The anemometry data should be available over some time period at the precise spot where any proposed WECS is to be built

➢ *Altitude of the proposed site.*

- It affects the air *density* and thus the power in the wind and hence the useful WECS electric power output
- The winds tend to have higher velocities at high altitudes.

➢ *Terrain and its aerodynamic*

- It may be possible to make use of hills or mountains which channel the prevailing winds into a pass region, thereby obtaining higher wind power

➢ *Local Ecology*

- If the surface is bare rock - lower hub heights hence lower structure cost.
- If trees or grass or vegetation are present, then higher hub heights - larger system costs than the bare ground case

➢ *Distance to Roads or Railways*

- For heavy machinery, structures, materials, blades and other apparatus will have to be moved into any chosen WECS site

➢ *Nearness of site to local centre/users*

- minimizes transmission line length and hence losses and costs

➢ *Nature of ground*

- foundations for a WECS are secured.
- Ground surface should be stable & free from erosion

➢ *Favourable land cost.*

- Land cost should be favourable